Mathematics for the Slightly Rusty but Still Ambitious

A Practical Review for Interviews, Work, and Future Moments of Panic

Erdan Mu Alan

University of Pennsylvania

Contents

Chapter 1

Linear Algebra

1.1 Systems of Linear Equations

Definition 1.1.1: Linear Equation

For $n \geq 1$, a *linear equation* in the variables $x_1, \ldots, x_n$ is an equation of the form

$$a_1x_1 + a_2x_2 + \cdots + a_nx_n = b$$

where b and the coefficients $a_1, \ldots, a_n$ are real (or complex) numbers. A *system of linear equations* (or *linear system*) is a collection of one or more linear equations involving the same variables.

Equations involving products, powers, or nonlinear functions of variables are not *linear. For example,* $x_1x_2 + 3x_2 = 4$ *and* $\sqrt{x_1} + x_2 = 1$ *are not linear equations.*

Definition 1.1.2: Solution and Solution Set

A *solution* of a linear system is an ordered list $(s_1, s_2, \ldots, s_n)$ of numbers that makes each equation a true statement when substituted for $x_1, \ldots, x_n$. The set of all solutions is called the *solution set.* Two linear systems are *equivalent* if they have the same solution set.

Consistency. A linear system is *consistent* if it has at least one solution (either exactly one or infinitely many). It is *inconsistent* if it has no solution. A consistent system has either exactly one solution or infinitely many solutions.

Definition 1.1.3: Matrix Notation

Given a linear system with m equations and n unknowns, the *coefficient matrix* is the $m \times n$ matrix of coefficients. The *augmented matrix* appends the column of constants to the right of the coefficient matrix, separated by a vertical bar.
For the system

$$\begin{cases} a_{11}x_1 + \cdots + a_{1n}x_n = b_1 \\ \vdots \\ a_{m1}x_1 + \cdots + a_{mn}x_n = b_m \end{cases}$$

the augmented matrix is

$$\begin{bmatrix} a_{11} & \cdots & a_{1n} & b_1 \\ \vdots & \ddots & \vdots & \vdots \\ a_{m1} & \cdots & a_{mn} & b_m \end{bmatrix}$$

Elementary Row Operations. Three operations produce equivalent systems:

1. *(Interchange)* Swap two rows: $R_i \leftrightarrow R_j$.
2. *(Scaling)* Multiply a row by a nonzero constant: $cR_i \to R_i$.
3. *(Replacement)* Replace one row by the sum of itself and a scalar multiple of another row: $R_i + cR_j \to R_i$.

Two matrices are *row equivalent* if one can be obtained from the other by a sequence of elementary row operations.

Definition 1.1.4: Echelon Forms

A matrix is in *(row) echelon form* (REF) if:

1. All nonzero rows are above any all-zero rows.
2. Each leading entry (leftmost nonzero entry) of a row lies strictly to the right of the leading entry of the row above it.
3. All entries below a leading entry in a column are zeros.

A matrix is in *reduced (row) echelon form* (RREF) if additionally:

4. Each leading entry is 1.
5. Each leading 1 is the only nonzero entry in its column.

Theorem 1.1.5: Uniqueness of RREF

Each matrix is row equivalent to one and only one reduced echelon matrix.

Definition 1.1.6: Pivot Positions

A *pivot position* in a matrix A is a location that corresponds to a leading 1 in the RREF of A. A *pivot column* is a column containing a pivot position. The nonzero number used to create zeros via row operations is called a *pivot*.

The Row Reduction Algorithm.

1. *(Forward phase)* Find the leftmost nonzero column; select a nonzero entry as pivot (interchange rows if needed); create zeros below the pivot using replacement; cover the pivot row and repeat on the remaining submatrix.

2. *(Backward phase)* Working from the rightmost pivot upward, scale each pivot row to make the pivot 1; use replacement to create zeros above each pivot.

The result is the RREF.

Basic and Free Variables. After row reducing the augmented matrix, variables corresponding to pivot columns are *basic variables*; all other variables are *free variables*. Free variables may take any value; basic variables are then determined uniquely.

Theorem 1.1.7: Existence and Uniqueness Theorem

A linear system is consistent if and only if the rightmost column of the augmented matrix is *not* a pivot column, i.e., no row of the form $[0 \ \cdots \ 0 \mid c]$ with $c \neq 0$ appears in the echelon form. If the system is consistent, it has:

- a *unique* solution if there are no free variables, or
- *infinitely many* solutions if there is at least one free variable.

Example 1.1.1

Solve the system

$$\begin{cases} x_1 - 2x_2 + x_3 = 0 \\ 2x_2 - 8x_3 = 8 \\ 5x_1 - 5x_3 = 10 \end{cases}$$

Row reducing the augmented matrix yields

$$\begin{bmatrix} 1 & -2 & 1 & 0 \\ 0 & 2 & -8 & 8 \\ 5 & 0 & -5 & 10 \end{bmatrix} \longrightarrow \begin{bmatrix} 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & -1 \end{bmatrix}$$

The unique solution is $x_1 = 1$, $x_2 = 0$, $x_3 = -1$.

1.2 Vector Equations and Matrix Equations

1.2.1 Vectors in $\mathbb{R}^n$

Definition 1.2.1: Vectors in $\mathbb{R}^n$

A vector in $\mathbb{R}^n$ is an $n \times 1$ column matrix

$$\mathbf{v} = \begin{bmatrix} v_1 \\ v_2 \\ \vdots \\ v_n \end{bmatrix}$$

The set $\mathbb{R}^n$ consists of all such n-dimensional column vectors. The *zero vector* $\mathbf{0}$ has all entries equal to zero.

Vector Operations.

- *Addition:* $\mathbf{u}+\mathbf{v} = \begin{bmatrix} u_1+v_1 \\ \vdots \\ u_n+v_n \end{bmatrix}$
- *Scalar multiplication:* $c\mathbf{u} = \begin{bmatrix} cu_1 \\ \vdots \\ cu_n \end{bmatrix}$

Proposition 1.2.2

(Algebraic properties of $\mathbb{R}^n$) For all $\mathbf{u}, \mathbf{v}, \mathbf{w} \in \mathbb{R}^n$ and scalars c, d:

$$\begin{aligned} &\mathbf{u}+\mathbf{v} = \mathbf{v}+\mathbf{u} && (\mathbf{u}+\mathbf{v})+\mathbf{w} = \mathbf{u}+(\mathbf{v}+\mathbf{w}) \\ &\mathbf{u}+\mathbf{0} = \mathbf{u} && \mathbf{u}+(-\mathbf{u}) = \mathbf{0} \\ &c(\mathbf{u}+\mathbf{v}) = c\mathbf{u}+c\mathbf{v} && (c+d)\mathbf{u} = c\mathbf{u}+d\mathbf{u} \\ &c(d\mathbf{u}) = (cd)\mathbf{u} && 1\mathbf{u} = \mathbf{u} \end{aligned}$$

1.2.2 Linear Combinations and Span

Definition 1.2.3: Linear Combination

Given vectors $\mathbf{v}_1, \ldots, \mathbf{v}_p \in \mathbb{R}^n$ and scalars $c_1, \ldots, c_p$, the vector

$$\mathbf{y} = c_1\mathbf{v}_1 + c_2\mathbf{v}_2 + \cdots + c_p\mathbf{v}_p$$

is called a *linear combination* of $\mathbf{v}_1, \ldots, \mathbf{v}_p$ with *weights* $c_1, \ldots, c_p$. The set of all linear combinations of $\mathbf{v}_1, \ldots, \mathbf{v}_p$ is denoted $\text{Span}\{\mathbf{v}_1, \ldots, \mathbf{v}_p\}$ and called the *subset of* $\mathbb{R}^n$ *spanned by* $\mathbf{v}_1, \ldots, \mathbf{v}_p$.

Remark.

> Asking whether $\mathbf{b}$ is in $\text{Span}\{\mathbf{v}_1, \ldots, \mathbf{v}_p\}$ is equivalent to asking whether the system with augmented matrix $[\mathbf{v}_1 \ \cdots \ \mathbf{v}_p \mid \mathbf{b}]$ is consistent.

1.2.3 The Matrix Equation $A\mathbf{x} = \mathbf{b}$

Definition 1.2.4: Matrix–Vector Product

If A is an $m \times n$ matrix with columns $\mathbf{a}_1, \ldots, \mathbf{a}_n$, and $\mathbf{x} \in \mathbb{R}^n$, then

$$A\mathbf{x} = x_1\mathbf{a}_1 + x_2\mathbf{a}_2 + \cdots + x_n\mathbf{a}_n = \sum_{i=1}^{n} x_i\mathbf{a}_i$$

That is, $A\mathbf{x}$ is a linear combination of the columns of A using the entries of $\mathbf{x}$ as weights. The product $A\mathbf{x}$ is defined only when the number of columns of A equals the number of entries in $\mathbf{x}$.

Proposition 1.2.5

The vector equation $x_1\mathbf{v}_1 + \cdots + x_p\mathbf{v}_p = \mathbf{b}$ has the same solution set as $A\mathbf{x} = \mathbf{b}$, where $A = [\mathbf{v}_1 \ \cdots \ \mathbf{v}_p]$.

Theorem 1.2.6: Spanning Criterion

Let A be an $m \times n$ matrix. The following are logically equivalent:

(a) For each $\mathbf{b} \in \mathbb{R}^m$, $A\mathbf{x} = \mathbf{b}$ has a solution.

(b) Each $\mathbf{b} \in \mathbb{R}^m$ is a linear combination of the columns of A.

(c) The columns of A span $\mathbb{R}^m$.

(d) A has a pivot position in every row.

Dot Product. For vectors $\mathbf{u}, \mathbf{v} \in \mathbb{R}^n$, the *dot product* (inner product) is

$$\mathbf{u} \cdot \mathbf{v} = \mathbf{u}^T\mathbf{v} = u_1v_1 + u_2v_2 + \cdots + u_nv_n$$

If A is $m \times n$ with rows $\mathbf{r}_1^T, \ldots, \mathbf{r}_m^T$, then the ith entry of $A\mathbf{x}$ equals $\mathbf{r}_i \cdot \mathbf{x}$.

1.3 Solution Sets of Linear Systems

1.3.1 Homogeneous Systems

Definition 1.3.1: Homogeneous Linear System

A linear system is *homogeneous* if it can be written as $A\mathbf{x} = \mathbf{0}$. Such a system always has the *trivial solution* $\mathbf{x} = \mathbf{0}$. A *nontrivial solution* is a nonzero vector satisfying $A\mathbf{x} = \mathbf{0}$.

Theorem 1.3.2: Nontrivial Solutions of Homogeneous Systems

The homogeneous equation $A\mathbf{x} = \mathbf{0}$ has a nontrivial solution if and only if the equation has at least one free variable.

Parametric Vector Form. When a homogeneous system $A\mathbf{x} = \mathbf{0}$ has free variables $x_{j_1}, \ldots, x_{j_k}$, the solution set can be written as

$$\mathbf{x} = x_{j_1}\mathbf{v}_1 + x_{j_2}\mathbf{v}_2 + \cdots + x_{j_k}\mathbf{v}_k$$

This is the *parametric vector form* of the solution. The solution set $\text{Span}\{\mathbf{v}_1, \ldots, \mathbf{v}_k\}$ is a subspace of $\mathbb{R}^n$.

Example 1.3.1

Describe the solution set of $3x_1 + 5x_2 - 4x_3 = 0$, $-3x_1 - 2x_2 + 4x_3 = 0$, $6x_1 + x_2 - 8x_3 = 0$. Row reduction gives

$$\begin{bmatrix} 3 & 5 & -4 \\ -3 & -2 & 4 \\ 6 & 1 & -8 \end{bmatrix} \longrightarrow \begin{bmatrix} 1 & 0 & -4/3 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

With x_3 free, $x_1 = \frac{4}{3}x_3$ and $x_2 = 0$. The solution set is $\text{Span}\left\{\begin{bmatrix} 4/3 \\ 0 \\ 1 \end{bmatrix}\right\}$, a line through the origin.

1.3.2 Nonhomogeneous Systems

Solution Structure. Suppose $A\mathbf{x} = \mathbf{b}$ is consistent and $\mathbf{p}$ is a particular solution. Then the solution set of $A\mathbf{x} = \mathbf{b}$ equals

$$\{\mathbf{p} + \mathbf{v}_h : \mathbf{v}_h \in \text{Nul}\, A\}$$

where $\text{Nul}\, A$ is the solution set of the associated homogeneous equation $A\mathbf{x} = \mathbf{0}$. Geometrically, the solution set is a translate of $\text{Nul}\, A$ by $\mathbf{p}$.

1.3.3 Linear Independence

Definition 1.3.3: Linear Independence

An indexed set $\{\mathbf{v}_1, \ldots, \mathbf{v}_p\}$ in $\mathbb{R}^n$ is *linearly independent* if the vector equation

$$x_1\mathbf{v}_1 + x_2\mathbf{v}_2 + \cdots + x_p\mathbf{v}_p = \mathbf{0}$$

has only the trivial solution $x_1 = x_2 = \cdots = x_p = 0$. The set is *linearly dependent* if there exist weights $c_1, \ldots, c_p$, not all zero, such that $c_1\mathbf{v}_1 + \cdots + c_p\mathbf{v}_p = \mathbf{0}$.

Proposition 1.3.4

The columns of a matrix A are linearly independent if and only if $A\mathbf{x} = \mathbf{0}$ has only the trivial solution.

Theorem 1.3.5: Linear Dependence Characterization

An indexed set $S = \{\mathbf{v}_1, \ldots, \mathbf{v}_p\}$ of two or more vectors is linearly dependent if and only if at least one of the vectors in S is a linear combination of the others. In particular, if S is linearly dependent and $\mathbf{v}_1 \neq \mathbf{0}$, then some $\mathbf{v}_j$ (with $j > 1$) is a linear combination of $\mathbf{v}_1, \ldots, \mathbf{v}_{j-1}$.

Theorem 1.3.6: Too Many Vectors

If a set $\{\mathbf{v}_1, \ldots, \mathbf{v}_p\} \subset \mathbb{R}^n$ has more vectors than entries ($p > n$), then the set is linearly dependent.

Proof

Let $A = [\mathbf{v}_1 \ \cdots \ \mathbf{v}_p]$. Since A is $n \times p$ with $p > n$, the equation $A\mathbf{x} = \mathbf{0}$ has more unknowns than equations, so there must be a free variable. Therefore $A\mathbf{x} = \mathbf{0}$ has a nontrivial solution and the columns are linearly dependent. $\square$

1.4 Linear Transformations

Definition 1.4.1: Transformation

A *transformation* (or *function* or *mapping*) $T : \mathbb{R}^n \to \mathbb{R}^m$ is a rule that assigns to each vector $\mathbf{x} \in \mathbb{R}^n$ a vector $T(\mathbf{x}) \in \mathbb{R}^m$. The set $\mathbb{R}^n$ is the *domain* and $\mathbb{R}^m$ is the *codomain*. The vector $T(\mathbf{x})$ is the *image* of $\mathbf{x}$, and the set of all images is the *range* of T.

Definition 1.4.2: Linear Transformation

A transformation $T : \mathbb{R}^n \to \mathbb{R}^m$ is *linear* if for all $\mathbf{u}, \mathbf{v} \in \mathbb{R}^n$ and all scalars c:

1. $T(\mathbf{u} + \mathbf{v}) = T(\mathbf{u}) + T(\mathbf{v})$
2. $T(c\mathbf{u}) = cT(\mathbf{u})$

Equivalently, $T(c\mathbf{u} + d\mathbf{v}) = cT(\mathbf{u}) + dT(\mathbf{v})$ for all $\mathbf{u}, \mathbf{v}$ and scalars c, d.

Every matrix transformation $\mathbf{x} \mapsto A\mathbf{x}$ is linear. Conversely, every linear transformation $T : \mathbb{R}^n \to \mathbb{R}^m$ is a matrix transformation.

Definition 1.4.3: Standard Matrix

Let $T : \mathbb{R}^n \to \mathbb{R}^m$ be a linear transformation. The *standard matrix* of T is the $m \times n$ matrix

$$A = [T(\mathbf{e}_1)\ T(\mathbf{e}_2)\ \cdots\ T(\mathbf{e}_n)]$$

where $\mathbf{e}_1, \ldots, \mathbf{e}_n$ are the standard basis vectors of $\mathbb{R}^n$. Then $T(\mathbf{x}) = A\mathbf{x}$ for all $\mathbf{x} \in \mathbb{R}^n$.

Geometric Transformations of $\mathbb{R}^2$. Common linear transformations with their standard matrices:

- *Reflection over the x_1-axis:* $A = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$
- *Rotation by angle θ:* $A = \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix}$
- *Projection onto x_1-axis:* $A = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$
- *Scaling by c:* $A = \begin{bmatrix} c & 0 \\ 0 & c \end{bmatrix}$

Theorem 1.4.4: One-to-One and Onto

Let $T : \mathbb{R}^n \to \mathbb{R}^m$ be a linear transformation with standard matrix A. Then:

1. T is *one-to-one* if and only if $A\mathbf{x} = \mathbf{0}$ has only the trivial solution (equivalently, the columns of A are linearly independent).
2. T maps $\mathbb{R}^n$ *onto* $\mathbb{R}^m$ if and only if the columns of A span $\mathbb{R}^m$ (equivalently, A has a pivot in every row).

1.5 Matrix Operations and Invertibility

1.5.1 Matrix Operations

Definition 1.5.1: Matrix Addition and Scalar Multiplication

If A and B are $m \times n$ matrices and r is a scalar, the *sum* $A+B$ is the $m \times n$ matrix with entries $(A+B)_{ij} = a_{ij}+b_{ij}$, and the *scalar multiple* rA has entries $(rA)_{ij} = ra_{ij}$.

Definition 1.5.2: Matrix Multiplication

If A is $m \times n$ and B is $n \times p$ with columns $\mathbf{b}_1, \ldots, \mathbf{b}_p$, the product AB is the $m \times p$ matrix

$$AB = [A\mathbf{b}_1 \; A\mathbf{b}_2 \; \cdots \; A\mathbf{b}_p]$$

The (i,j)-entry of AB is $\sum_{k=1}^{n} a_{ik}b_{kj}$ (dot product of the ith row of A with the jth column of B).

Remark.

> Matrix multiplication corresponds to the composition of linear transformations: if $T(\mathbf{x}) = A\mathbf{x}$ and $S(\mathbf{x}) = B\mathbf{x}$, then $(T \circ S)(\mathbf{x}) = A(B\mathbf{x}) = (AB)\mathbf{x}$.

Proposition 1.5.3

(Properties of matrix operations) Let A, B, C be matrices of compatible sizes and r, s scalars:

$$A(BC) = (AB)C \qquad A(B+C) = AB + AC$$
$$(B+C)A = BA + CA \qquad r(AB) = (rA)B = A(rB)$$
$$(AB)^T = B^T A^T$$

In general, $AB \neq BA$; matrix multiplication is *not* commutative.

Definition 1.5.4: Transpose

The *transpose* of an $m \times n$ matrix A is the $n \times m$ matrix A^T whose (i,j)-entry is the (j,i)-entry of A. The rows of A become the columns of A^T.

Definition 1.5.5: Identity Matrix

The $n \times n$ *identity matrix* I_n has 1s on the main diagonal and 0s elsewhere: $(I_n)_{ij} = 1$ if $i = j$ and 0 otherwise. It satisfies $AI_n = A$ and $I_m A = A$ for any $m \times n$ matrix A.

1.5.2 The Inverse of a Matrix

Definition 1.5.6: Invertible Matrix

An $n \times n$ matrix A is *invertible* (or *nonsingular*) if there exists an $n \times n$ matrix A^{-1} such that

$$A^{-1}A = I_n \quad \text{and} \quad AA^{-1} = I_n$$

A matrix that is not invertible is called *singular*.

Theorem 1.5.7: 2×2 Inverse Formula

Let $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$. If $ad - bc \neq 0$, then A is invertible and

$$A^{-1} = \frac{1}{ad - bc} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$$

If $ad - bc = 0$, then A is not invertible. The quantity $\det A = ad - bc$ is called the *determinant* of A.

Theorem 1.5.8: Properties of Invertible Matrices

(a) If A is invertible, so is A^{-1}, and $(A^{-1})^{-1} = A$.

(b) If A and B are invertible $n \times n$ matrices, so is AB, and $(AB)^{-1} = B^{-1}A^{-1}$.

(c) If A is invertible, so is A^T, and $(A^T)^{-1} = (A^{-1})^T$.

Proof

(b): Compute $(AB)(B^{-1}A^{-1}) = A(BB^{-1})A^{-1} = AIA^{-1} = AA^{-1} = I$. Similarly $(B^{-1}A^{-1})(AB) = I$. (c): $(A^{-1})^T A^T = (AA^{-1})^T = I^T = I$, and $A^T(A^{-1})^T = (A^{-1}A)^T = I$. $\square$

Algorithm for Finding A^{-1}. Row reduce the augmented matrix $[A \mid I]$. If A is row equivalent to I, then $[A \mid I]$ is row equivalent to $[I \mid A^{-1}]$. Otherwise A is not invertible.

Theorem 1.5.9: The Invertible Matrix Theorem

Let A be an $n \times n$ matrix. The following statements are all equivalent:

(a) A is invertible.

(b) A is row equivalent to I_n.

(c) A has n pivot positions.

(d) $A\mathbf{x} = \mathbf{0}$ has only the trivial solution.

(e) The columns of A form a linearly independent set.

(f) The linear transformation $\mathbf{x} \mapsto A\mathbf{x}$ is one-to-one.

(g) $A\mathbf{x} = \mathbf{b}$ has exactly one solution for each $\mathbf{b} \in \mathbb{R}^n$.

(h) The columns of A span $\mathbb{R}^n$.

(i) $\mathbf{x} \mapsto A\mathbf{x}$ maps $\mathbb{R}^n$ onto $\mathbb{R}^n$.

(j) There is an $n \times n$ matrix C with $CA = I$.

(k) There is an $n \times n$ matrix D with $AD = I$.

(l) A^T is invertible.

1.6 Determinants

1.6.1 Definition via Cofactor Expansion

Definition 1.6.1: Cofactor Expansion

For an $n \times n$ matrix A, the (i, j)-*cofactor* is

$$C_{ij} = (-1)^{i+j} \det A_{ij}$$

where A_{ij} is the $(n-1) \times (n-1)$ submatrix obtained by deleting row i and column j from A. The *determinant* of A is defined recursively by expansion along any row or column:

$$\det A = a_{i1}C_{i1} + a_{i2}C_{i2} + \cdots + a_{in}C_{in} \quad \text{(expansion along row } i\text{)}$$

For a 1×1 matrix, $\det[a] = a$. For a 2×2 matrix, $\det \begin{bmatrix} a & b \\ c & d \end{bmatrix} = ad - bc$.

Example 1.6.1

Rule of Sarrus for 3×3 matrices: given $A = [a_{ij}]$,

$$\det A = a_{11}a_{22}a_{33} + a_{12}a_{23}a_{31} + a_{13}a_{21}a_{32} - a_{11}a_{23}a_{32} - a_{12}a_{21}a_{33} - a_{13}a_{22}a_{31}$$

which equals $a_{11}(a_{22}a_{33} - a_{23}a_{32}) - a_{12}(a_{21}a_{33} - a_{23}a_{31}) + a_{13}(a_{21}a_{32} - a_{22}a_{31})$.

1.6.2 Properties of Determinants

Theorem 1.6.2: Row Operations and Determinants

Let A be an $n \times n$ matrix.

(a) *(Replacement)* If a multiple of one row is added to another row, the determinant is unchanged.

(b) *(Interchange)* Swapping two rows multiplies the determinant by -1.

(c) *(Scaling)* Multiplying a row by a scalar k multiplies the determinant by k.

Proposition 1.6.3

$\det A = 0$ if and only if A is not invertible (singular). Equivalently, A is invertible if and only if $\det A \neq 0$.

Theorem 1.6.4: Multiplicative Property

If A and B are $n \times n$ matrices, then $\det(AB) = (\det A)(\det B)$.

Proposition 1.6.5

$\det A^T = \det A$.

1.6.3 Cramer's Rule and the Adjugate

Theorem 1.6.6: Cramer's Rule

Let A be an invertible $n \times n$ matrix. For any $\mathbf{b} \in \mathbb{R}^n$, the unique solution $\mathbf{x}$ of $A\mathbf{x} = \mathbf{b}$ has entries

$$x_i = \frac{\det A_i(\mathbf{b})}{\det A}, \quad i = 1, 2, \ldots, n$$

where $A_i(\mathbf{b})$ is the matrix obtained from A by replacing the ith column by $\mathbf{b}$.

Definition 1.6.7: Adjugate

The *adjugate* (or *classical adjoint*) of A is the transpose of the cofactor matrix:

$$\operatorname{adj} A = (C_{ij})^T, \quad \text{so } (\operatorname{adj} A)_{ij} = C_{ji}$$

Theorem 1.6.8: Adjugate Formula for the Inverse

If A is invertible, then $A^{-1} = \dfrac{1}{\det A} \operatorname{adj} A$.

1.6.4 Geometric Interpretation

Theorem 1.6.9: Determinant as Area and Volume

If A is a 2×2 matrix, the area of the parallelogram determined by the columns of A is $|\det A|$. If A is a 3×3 matrix, the volume of the parallelepiped determined by the columns of A is $|\det A|$.
More generally, if $T : \mathbb{R}^2 \to \mathbb{R}^2$ is determined by a 2×2 matrix A and S is a region in $\mathbb{R}^2$, then $\{\text{area of } T(S)\} = |\det A| \cdot \{\text{area of } S\}$.

1.7 Vector Spaces and Subspaces

1.7.1 Vector Spaces

Definition 1.7.1: Vector Space

A *vector space* is a nonempty set V together with two operations—addition $+ : V \times V \to V$ and scalar multiplication $\cdot : \mathbb{R} \times V \to V$—satisfying the following ten axioms for all $\mathbf{u}, \mathbf{v}, \mathbf{w} \in V$ and all scalars c, d:

1. $\mathbf{u} + \mathbf{v} \in V$
2. $\mathbf{u} + \mathbf{v} = \mathbf{v} + \mathbf{u}$
3. $(\mathbf{u} + \mathbf{v}) + \mathbf{w} = \mathbf{u} + (\mathbf{v} + \mathbf{w})$
4. There exists $\mathbf{0} \in V$ such that $\mathbf{u} + \mathbf{0} = \mathbf{u}$
5. For each $\mathbf{u}$ there exists $-\mathbf{u}$ with $\mathbf{u} + (-\mathbf{u}) = \mathbf{0}$
6. $c\mathbf{u} \in V$
7. $c(\mathbf{u} + \mathbf{v}) = c\mathbf{u} + c\mathbf{v}$
8. $(c + d)\mathbf{u} = c\mathbf{u} + d\mathbf{u}$
9. $c(d\mathbf{u}) = (cd)\mathbf{u}$
10. $1\mathbf{u} = \mathbf{u}$

Example 1.7.1

Examples of vector spaces:

- $\mathbb{R}^n$ with standard addition and scalar multiplication.
- $\mathcal{P}_n$: polynomials of degree at most n, with pointwise addition and scalar multiplication.
- $\mathcal{P}$: all polynomials (infinite-dimensional).
- $M_{m \times n}$: all $m \times n$ real matrices.
- The space of all real-valued continuous functions on an interval.

Proposition 1.7.2

For each $\mathbf{u} \in V$ and scalar c: $0\mathbf{u} = \mathbf{0}$, $c\mathbf{0} = \mathbf{0}$, $(-1)\mathbf{u} = -\mathbf{u}$.

1.7.2 Subspaces

Definition 1.7.3: Subspace

A *subspace* of a vector space V is a subset $H \subseteq V$ satisfying:

(a) $\mathbf{0} \in H$.

(b) For all $\mathbf{u}, \mathbf{v} \in H$: $\mathbf{u} + \mathbf{v} \in H$ (closed under addition).

(c) For all $\mathbf{u} \in H$ and scalars c: $c\mathbf{u} \in H$ (closed under scalar multiplication).

A subspace is itself a vector space under the same operations.

Theorem 1.7.4: Span is a Subspace

If $\mathbf{v}_1, \ldots, \mathbf{v}_p$ are in a vector space V, then $\text{Span}\{\mathbf{v}_1, \ldots, \mathbf{v}_p\}$ is a subspace of V.

1.7.3 Null Space, Column Space, and Row Space

Definition 1.7.5: Null Space

The *null space* of an $m \times n$ matrix A, written $\operatorname{Nul} A$, is the set of all solutions of $A\mathbf{x} = \mathbf{0}$:

$$\operatorname{Nul} A = \{\mathbf{x} \in \mathbb{R}^n : A\mathbf{x} = \mathbf{0}\}$$

Theorem 1.7.6: Null Space is a Subspace

$\operatorname{Nul} A$ is a subspace of $\mathbb{R}^n$.

Proof

$A\mathbf{0} = \mathbf{0}$, so $\mathbf{0} \in \operatorname{Nul} A$. If $\mathbf{u}, \mathbf{v} \in \operatorname{Nul} A$ then $A(\mathbf{u} + \mathbf{v}) = A\mathbf{u} + A\mathbf{v} = \mathbf{0}$, so $\mathbf{u} + \mathbf{v} \in \operatorname{Nul} A$. If $\mathbf{u} \in \operatorname{Nul} A$ then $A(c\mathbf{u}) = cA\mathbf{u} = \mathbf{0}$, so $c\mathbf{u} \in \operatorname{Nul} A$. □

Definition 1.7.7: Column Space

The *column space* of an $m \times n$ matrix A, written $\operatorname{Col} A$, is the set of all linear combinations of the columns of A:

$$\operatorname{Col} A = \text{Span}\{\mathbf{a}_1, \ldots, \mathbf{a}_n\} = \{A\mathbf{x} : \mathbf{x} \in \mathbb{R}^n\}$$

This is a subspace of $\mathbb{R}^m$. The equation $A\mathbf{x} = \mathbf{b}$ is consistent if and only if $\mathbf{b} \in \operatorname{Col} A$.

Definition 1.7.8: Row Space

The *row space* of an $m \times n$ matrix A, written $\operatorname{Row} A$, is the subspace of $\mathbb{R}^n$ spanned by the rows of A. Note that $\operatorname{Row} A = \operatorname{Col} A^T$.

Contrast Between Nul A and Col A. For an $m \times n$ matrix A: $\operatorname{Nul} A$ is a subspace of $\mathbb{R}^n$ (involving columns), while $\operatorname{Col} A$ is a subspace of $\mathbb{R}^m$ (involving rows in the range).

1.7.4 Bases

Definition 1.7.9: Basis

A *basis* for a vector space V is a linearly independent set $\mathcal{B} = \{\mathbf{b}_1, \dots, \mathbf{b}_n\}$ that spans V.

Example 1.7.2

The *standard basis* for $\mathbb{R}^n$ is $\{\mathbf{e}_1, \dots, \mathbf{e}_n\}$, the columns of I_n. The standard basis for $\mathcal{P}_n$ is $\{1, t, t^2, \dots, t^n\}$.

Bases for Nul A and Col A.

- A basis for $\operatorname{Nul} A$: solve $A\mathbf{x} = \mathbf{0}$ in parametric vector form; the spanning vectors form a basis.
- A basis for $\operatorname{Col} A$: the pivot columns of A form a basis for $\operatorname{Col} A$.
- A basis for $\operatorname{Row} A$: the nonzero rows of any echelon form of A form a basis for $\operatorname{Row} A$.

Theorem 1.7.10: Spanning Set Theorem

Let $S = \{\mathbf{v}_1, \dots, \mathbf{v}_p\}$ be a set in a vector space V with $H = \operatorname{Span} S$.

(a) If one vector in S, say $\mathbf{v}_k$, is a linear combination of the others, then $\operatorname{Span}\{S \setminus \{\mathbf{v}_k\}\} = H$.

(b) If $H \neq \{\mathbf{0}\}$, some subset of S is a basis for H.

1.8 Coordinate Systems and Dimension

1.8.1 Coordinate Vectors

Theorem 1.8.1: Unique Representation Theorem

Let $\mathcal{B} = \{\mathbf{b}_1, \dots, \mathbf{b}_n\}$ be a basis for a vector space V. Then for each $\mathbf{x} \in V$, there exists a unique set of scalars $c_1, \dots, c_n$ such that

$$\mathbf{x} = c_1\mathbf{b}_1 + c_2\mathbf{b}_2 + \cdots + c_n\mathbf{b}_n$$

Definition 1.8.2: Coordinate Vector

Suppose $\mathcal{B} = \{\mathbf{b}_1, \dots, \mathbf{b}_n\}$ is a basis for V and $\mathbf{x} = c_1\mathbf{b}_1 + \cdots + c_n\mathbf{b}_n$. The $\mathcal{B}$-*coordinate vector* of $\mathbf{x}$ is

$$[\mathbf{x}]_{\mathcal{B}} = \begin{bmatrix} c_1 \\ \vdots \\ c_n \end{bmatrix} \in \mathbb{R}^n$$

The *change-of-coordinates matrix* from $\mathcal{B}$ to the standard basis is $P_{\mathcal{B}} = [\mathbf{b}_1 \ \cdots \ \mathbf{b}_n]$, and $\mathbf{x} = P_{\mathcal{B}}[\mathbf{x}]_{\mathcal{B}}$ (for $V = \mathbb{R}^n$).

Theorem 1.8.3: Coordinate Mapping is an Isomorphism

If $\mathcal{B} = \{\mathbf{b}_1, \dots, \mathbf{b}_n\}$ is a basis for V, then the coordinate mapping $\mathbf{x} \mapsto [\mathbf{x}]_{\mathcal{B}}$ is a one-to-one linear transformation from V onto $\mathbb{R}^n$.

1.8.2 Dimension

Theorem 1.8.4: Basis Size is Unique

If a vector space V has a basis of n vectors, then every basis of V has exactly n vectors.

Proof

Let $\mathcal{B}_1$ be a basis with n vectors and $\mathcal{B}_2$ any other basis. Since $\mathcal{B}_1$ is a basis and $\mathcal{B}_2$ is linearly independent, $|\mathcal{B}_2| \leq n$. By symmetry $|\mathcal{B}_2| \geq n$. Hence $|\mathcal{B}_2| = n$. □

Definition 1.8.5: Dimension

The *dimension* of a nonzero finite-dimensional vector space V, written $\dim V$, is the number of vectors in any basis for V. The dimension of $\{\mathbf{0}\}$ is defined to be 0. If V cannot be spanned by a finite set, it is *infinite-dimensional*.

Example 1.8.1

$\dim \mathbb{R}^n = n$. $\dim \mathcal{P}_n = n + 1$. $\dim M_{m\times n} = mn$. $\dim \mathcal{P} = \infty$.

Theorem 1.8.6: The Basis Theorem

Let V be an n-dimensional vector space, $n \geq 1$. Any linearly independent set of exactly n elements in V is automatically a basis for V. Any set of exactly n elements that spans V is automatically a basis for V.

1.8.3 Rank and Nullity

Definition 1.8.7: Rank and Nullity

The *rank* of a matrix A, written $\operatorname{rank} A$, is the dimension of $\operatorname{Col} A$ (equal to the number of pivot columns). The *nullity* of A is $\dim \operatorname{Nul} A$ (equal to the number of free variables in $A\mathbf{x} = \mathbf{0}$).

Theorem 1.8.8: The Rank Theorem

If A is an $m \times n$ matrix, then

$$\operatorname{rank} A + \operatorname{nullity} A = n$$

Theorem 1.8.9: The Rank-Nullity Theorem

Let V, W be finite-dimensional vector spaces and $T : V \to W$ a linear transformation. Then

$$\dim(\ker T) + \dim(\operatorname{Im} T) = \dim V$$

where $\ker T = \{\mathbf{x} \in V : T(\mathbf{x}) = \mathbf{0}\}$ is the kernel and $\operatorname{Im} T = \{T(\mathbf{x}) : \mathbf{x} \in V\}$ is the image of T.

Extended Invertible Matrix Theorem. For an $n \times n$ matrix A, the following are also equivalent to the statements in the Invertible Matrix Theorem (1.5.9):

- The columns of A form a basis of $\mathbb{R}^n$.
- $\operatorname{Col} A = \mathbb{R}^n$.
- $\operatorname{rank} A = n$.
- $\operatorname{nullity} A = 0$.
- $\operatorname{Nul} A = \{\mathbf{0}\}$.

1.9 Eigenvalues and Eigenvectors

1.9.1 Definitions and the Characteristic Equation

Definition 1.9.1: Eigenvalue and Eigenvector

An *eigenvector* of an $n \times n$ matrix A is a *nonzero* vector $\mathbf{x}$ such that $A\mathbf{x} = \lambda\mathbf{x}$ for some scalar λ. Such a scalar λ is called an *eigenvalue* of A. The set of all solutions to $A\mathbf{x} = \lambda\mathbf{x}$, including $\mathbf{0}$, is the *eigenspace* $E_\lambda = \operatorname{Nul}(A - \lambda I)$.

Remark.

> λ is an eigenvalue of A if and only if $(A - \lambda I)\mathbf{x} = \mathbf{0}$ has a nontrivial solution, i.e., if and only if $\det(A - \lambda I) = 0$.

Definition 1.9.2: Characteristic Equation

The *characteristic polynomial* of A is $p(\lambda) = \det(A - \lambda I)$. The equation

$$\det(A - \lambda I) = 0$$

is the *characteristic equation* of A. Its roots are the eigenvalues of A.

Example 1.9.1

Find the eigenvalues of $A = \begin{bmatrix} 3 & -2 \\ 1 & 0 \end{bmatrix}$.

$$\det(A - \lambda I) = \begin{vmatrix} 3-\lambda & -2 \\ 1 & -\lambda \end{vmatrix} = -\lambda(3-\lambda) + 2 = \lambda^2 - 3\lambda + 2 = (\lambda - 1)(\lambda - 2) = 0$$

The eigenvalues are $\lambda_1 = 1$ and $\lambda_2 = 2$.

Theorem 1.9.3: Eigenvectors from Distinct Eigenvalues are Independent

If $\mathbf{v}_1, \ldots, \mathbf{v}_r$ are eigenvectors corresponding to distinct eigenvalues $\lambda_1, \ldots, \lambda_r$ of a matrix A, then $\{\mathbf{v}_1, \ldots, \mathbf{v}_r\}$ is linearly independent.

Eigenvalues of Triangular Matrices. The eigenvalues of a triangular matrix are its diagonal entries.

1.9.2 Diagonalization

Definition 1.9.4: Diagonalizable Matrix

An $n \times n$ matrix A is *diagonalizable* if $A = PDP^{-1}$ for some invertible matrix P and diagonal matrix D. The diagonal entries of D are eigenvalues of A and the columns of P are corresponding eigenvectors.

Theorem 1.9.5: The Diagonalization Theorem

An $n \times n$ matrix A is diagonalizable if and only if A has n linearly independent eigenvectors. In fact, $A = PDP^{-1}$ with D diagonal if and only if the columns of P are n linearly independent eigenvectors of A, and the diagonal entries of D are the corresponding eigenvalues.

Proof

($\Rightarrow$) If $A = PDP^{-1}$ then $AP = PD$. Writing $P = [\mathbf{v}_1 \cdots \mathbf{v}_n]$ and $D = \mathrm{diag}(\lambda_1, \ldots, \lambda_n)$, the jth column of $AP = PD$ gives $A\mathbf{v}_j = \lambda_j \mathbf{v}_j$. Since P is invertible, the $\mathbf{v}_j$ are nonzero and linearly independent.
($\Leftarrow$) If A has n linearly independent eigenvectors $\mathbf{v}_j$ with eigenvalues λ_j, set $P = [\mathbf{v}_1 \cdots \mathbf{v}_n]$ and $D = \mathrm{diag}(\lambda_1, \ldots, \lambda_n)$. Then $AP = PD$ and P is invertible, giving $A = PDP^{-1}$. $\square$

Algorithm: Diagonalizing A.

1. Find all eigenvalues by solving $\det(A - \lambda I) = 0$.
2. For each eigenvalue, find a basis for the eigenspace $\mathrm{Nul}(A - \lambda I)$.
3. A is diagonalizable if and only if the total number of basis vectors equals n.
4. Form P with eigenvectors as columns, and D with corresponding eigenvalues on the diagonal.

Corollary 1.9.6

If an $n \times n$ matrix A has n distinct eigenvalues, then A is diagonalizable.

1.9.3 Complex Eigenvalues

Remark.

The characteristic polynomial may have complex roots. A real 2×2 matrix with no real eigenvalues has a characteristic polynomial $\lambda^2 - (\operatorname{tr} A)\lambda + \det A = 0$ with complex roots $\lambda = a \pm bi$ (with $b \neq 0$).
For a real matrix with eigenvalue $\lambda = a + bi$ and complex eigenvector $\mathbf{v} = \mathbf{p} + i\mathbf{q}$ (with $\mathbf{p}, \mathbf{q} \in \mathbb{R}^n$), the real canonical form provides a block diagonal decomposition with 2×2 blocks $\begin{bmatrix} a & -b \\ b & a \end{bmatrix}$ representing rotation-scaling.

1.10 Inner Products and Orthogonality

1.10.1 The Inner Product in $\mathbb{R}^n$

Definition 1.10.1: Inner Product (Dot Product) in $\mathbb{R}^n$

For $\mathbf{u}, \mathbf{v} \in \mathbb{R}^n$, the *inner product* (or *dot product*) is

$$\mathbf{u} \cdot \mathbf{v} = \mathbf{u}^T \mathbf{v} = u_1 v_1 + u_2 v_2 + \cdots + u_n v_n$$

Proposition 1.10.2

For $\mathbf{u}, \mathbf{v}, \mathbf{w} \in \mathbb{R}^n$ and scalar c:

(a) $\mathbf{u} \cdot \mathbf{v} = \mathbf{v} \cdot \mathbf{u}$

(b) $(\mathbf{u} + \mathbf{v}) \cdot \mathbf{w} = \mathbf{u} \cdot \mathbf{w} + \mathbf{v} \cdot \mathbf{w}$

(c) $(c\mathbf{u}) \cdot \mathbf{v} = c(\mathbf{u} \cdot \mathbf{v}) = \mathbf{u} \cdot (c\mathbf{v})$

(d) $\mathbf{u} \cdot \mathbf{u} \geq 0$, and $\mathbf{u} \cdot \mathbf{u} = 0$ if and only if $\mathbf{u} = \mathbf{0}$.

Definition 1.10.3: Norm and Unit Vector

The *norm* (length) of $\mathbf{v} \in \mathbb{R}^n$ is $\|\mathbf{v}\| = \sqrt{\mathbf{v} \cdot \mathbf{v}} = \sqrt{v_1^2 + \cdots + v_n^2}$. A *unit vector* has norm 1. To *normalize* $\mathbf{v} \neq \mathbf{0}$, compute $\mathbf{u} = \mathbf{v} / \|\mathbf{v}\|$.

Definition 1.10.4: Orthogonality

Two vectors $\mathbf{u}, \mathbf{v} \in \mathbb{R}^n$ are *orthogonal* if $\mathbf{u} \cdot \mathbf{v} = 0$. The *orthogonal complement* of a subspace W is

$$W^\perp = \{\mathbf{z} \in \mathbb{R}^n : \mathbf{z} \cdot \mathbf{w} = 0 \text{ for all } \mathbf{w} \in W\}$$

Theorem 1.10.5: Fundamental Subspace Relationship

For any $m \times n$ matrix A:

$$(\operatorname{Col} A)^\perp = \operatorname{Nul} A^T \quad \text{and} \quad (\operatorname{Row} A)^\perp = \operatorname{Nul} A$$

1.10.2 Orthogonal Sets and Projections

Definition 1.10.6: Orthogonal Set and Orthogonal Basis

A set $\{\mathbf{u}_1, \dots, \mathbf{u}_p\}$ in $\mathbb{R}^n$ is an *orthogonal set* if $\mathbf{u}_i \cdot \mathbf{u}_j = 0$ for all $i \neq j$. If each $\mathbf{u}_i$ also satisfies $\|\mathbf{u}_i\| = 1$, the set is *orthonormal*. An orthogonal (respectively orthonormal) set that is a basis for a subspace W is an *orthogonal* (resp. *orthonormal*) *basis* for W.

Theorem 1.10.7: Orthogonal Sets are Linearly Independent

If $\{\mathbf{u}_1, \dots, \mathbf{u}_p\}$ is an orthogonal set of nonzero vectors, then it is linearly independent.

Proof

Suppose $\sum_{i=1}^p c_i \mathbf{u}_i = \mathbf{0}$. For fixed j, take the dot product with $\mathbf{u}_j$: $c_j \|\mathbf{u}_j\|^2 = 0$. Since $\mathbf{u}_j \neq \mathbf{0}$, $c_j = 0$. □

Theorem 1.10.8: Coordinates from an Orthogonal Basis

If $\{\mathbf{u}_1, \dots, \mathbf{u}_p\}$ is an orthogonal basis for a subspace W, then for each $\mathbf{y} \in W$:

$$\mathbf{y} = \frac{\mathbf{y} \cdot \mathbf{u}_1}{\mathbf{u}_1 \cdot \mathbf{u}_1} \mathbf{u}_1 + \frac{\mathbf{y} \cdot \mathbf{u}_2}{\mathbf{u}_2 \cdot \mathbf{u}_2} \mathbf{u}_2 + \cdots + \frac{\mathbf{y} \cdot \mathbf{u}_p}{\mathbf{u}_p \cdot \mathbf{u}_p} \mathbf{u}_p$$

Theorem 1.10.9: Orthogonal Decomposition Theorem

Let W be a subspace of $\mathbb{R}^n$. Every $\mathbf{y} \in \mathbb{R}^n$ can be written uniquely as

$$\mathbf{y} = \hat{\mathbf{y}} + \mathbf{z}$$

where $\hat{\mathbf{y}} \in W$ and $\mathbf{z} \in W^\perp$. If $\{\mathbf{u}_1, \dots, \mathbf{u}_p\}$ is any orthogonal basis for W, then

$$\hat{\mathbf{y}} = \operatorname{proj}_W \mathbf{y} = \frac{\mathbf{y} \cdot \mathbf{u}_1}{\mathbf{u}_1 \cdot \mathbf{u}_1} \mathbf{u}_1 + \cdots + \frac{\mathbf{y} \cdot \mathbf{u}_p}{\mathbf{u}_p \cdot \mathbf{u}_p} \mathbf{u}_p$$

and $\mathbf{z} = \mathbf{y} - \hat{\mathbf{y}}$.

Theorem 1.10.10: The Best Approximation Theorem

Let W be a subspace of $\mathbb{R}^n$, $\mathbf{y} \in \mathbb{R}^n$, and $\hat{\mathbf{y}} = \operatorname{proj}_W \mathbf{y}$. Then $\hat{\mathbf{y}}$ is the closest point in W to $\mathbf{y}$:

$$\|\mathbf{y} - \hat{\mathbf{y}}\| < \|\mathbf{y} - \mathbf{v}\| \quad \text{for all } \mathbf{v} \in W,\ \mathbf{v} \neq \hat{\mathbf{y}}$$

1.10.3 Orthonormal Matrices

Theorem 1.10.11: Orthonormal Columns

An $m \times n$ matrix U has orthonormal columns if and only if $U^T U = I_n$. If U has orthonormal columns, then $\|U\mathbf{x}\| = \|\mathbf{x}\|$ and $(U\mathbf{x}) \cdot (U\mathbf{y}) = \mathbf{x} \cdot \mathbf{y}$ for all $\mathbf{x}, \mathbf{y} \in \mathbb{R}^n$.

Definition 1.10.12: Orthogonal Matrix

A square $n \times n$ matrix U is an *orthogonal matrix* if $U^{-1} = U^T$, i.e., $UU^T = U^T U = I_n$. Equivalently, U has orthonormal columns (and rows).

Theorem 1.10.13: Projection via Orthonormal Basis

If $\{\mathbf{u}_1, \ldots, \mathbf{u}_p\}$ is an orthonormal basis for a subspace W and $U = [\mathbf{u}_1 \ \cdots \ \mathbf{u}_p]$, then

$$\operatorname{proj}_W \mathbf{y} = (\mathbf{y} \cdot \mathbf{u}_1)\mathbf{u}_1 + \cdots + (\mathbf{y} \cdot \mathbf{u}_p)\mathbf{u}_p = UU^T \mathbf{y}$$

1.10.4 The Gram-Schmidt Process

Theorem 1.10.14: Gram-Schmidt Process

Given a basis $\{\mathbf{x}_1, \ldots, \mathbf{x}_p\}$ for a nonzero subspace W of $\mathbb{R}^n$, define

$$\begin{aligned} \mathbf{v}_1 &= \mathbf{x}_1 \\ \mathbf{v}_k &= \mathbf{x}_k - \sum_{j=1}^{k-1} \frac{\mathbf{x}_k \cdot \mathbf{v}_j}{\mathbf{v}_j \cdot \mathbf{v}_j} \mathbf{v}_j \quad (k = 2, \ldots, p) \end{aligned}$$

Then $\{\mathbf{v}_1, \ldots, \mathbf{v}_p\}$ is an orthogonal basis for W, and $\operatorname{Span}\{\mathbf{v}_1, \ldots, \mathbf{v}_k\} = \operatorname{Span}\{\mathbf{x}_1, \ldots, \mathbf{x}_k\}$ for each k. An orthonormal basis is obtained by normalizing: $\mathbf{u}_k = \mathbf{v}_k / \|\mathbf{v}_k\|$.

Example 1.10.1

Apply Gram-Schmidt to $\mathbf{x}_1 = \begin{bmatrix}1\\1\\1\\1\end{bmatrix}$, $\mathbf{x}_2 = \begin{bmatrix}0\\1\\1\\1\end{bmatrix}$:

$$\mathbf{v}_1 = \begin{bmatrix}1\\1\\1\\1\end{bmatrix}, \quad \mathbf{v}_2 = \mathbf{x}_2 - \frac{\mathbf{x}_2 \cdot \mathbf{v}_1}{\mathbf{v}_1 \cdot \mathbf{v}_1}\mathbf{v}_1 = \begin{bmatrix}0\\1\\1\\1\end{bmatrix} - \frac{3}{4}\begin{bmatrix}1\\1\\1\\1\end{bmatrix} = \begin{bmatrix}-3/4\\1/4\\1/4\\1/4\end{bmatrix}$$

These form an orthogonal basis for $W = \text{Span}\{\mathbf{x}_1, \mathbf{x}_2\}$.

1.10.5 QR Factorization

Theorem 1.10.15: QR Factorization

If A is an $m \times n$ matrix with linearly independent columns, then $A = QR$, where Q is an $m \times n$ matrix with orthonormal columns that form a basis for $\operatorname{Col} A$, and R is an $n \times n$ upper triangular matrix with positive diagonal entries. This factorization is unique.

Algorithm for QR Factorization.

1. Apply the Gram-Schmidt process (with normalization) to the columns $\mathbf{x}_1, \dots, \mathbf{x}_n$ of A to obtain an orthonormal basis $\mathbf{u}_1, \dots, \mathbf{u}_n$ for $\operatorname{Col} A$.
2. Set $Q = [\mathbf{u}_1 \ \cdots \ \mathbf{u}_n]$.
3. Compute $R = Q^T A$ (an upper triangular matrix with positive diagonal entries).

1.10.6 Inner Product Spaces

Definition 1.10.16: Inner Product Space

An *inner product* on a real vector space V is a function $\langle \cdot, \cdot \rangle : V \times V \to \mathbb{R}$ satisfying for all $\mathbf{u}, \mathbf{v}, \mathbf{w} \in V$ and scalars c:

1. $\langle \mathbf{u}, \mathbf{v} \rangle = \langle \mathbf{v}, \mathbf{u} \rangle$
2. $\langle \mathbf{u} + \mathbf{v}, \mathbf{w} \rangle = \langle \mathbf{u}, \mathbf{w} \rangle + \langle \mathbf{v}, \mathbf{w} \rangle$
3. $\langle c\mathbf{u}, \mathbf{v} \rangle = c\langle \mathbf{u}, \mathbf{v} \rangle$
4. $\langle \mathbf{u}, \mathbf{u} \rangle \geq 0$, and $\langle \mathbf{u}, \mathbf{u} \rangle = 0$ iff $\mathbf{u} = \mathbf{0}$

A vector space with an inner product is an *inner product space*. The induced norm is $\|\mathbf{u}\| = \sqrt{\langle \mathbf{u}, \mathbf{u} \rangle}$.

Theorem 1.10.17: Cauchy–Schwarz Inequality

In any inner product space, for all $\mathbf{u}, \mathbf{v}$:

$$|\langle \mathbf{u}, \mathbf{v} \rangle| \leq \|\mathbf{u}\| \, \|\mathbf{v}\|$$

with equality if and only if $\mathbf{u}$ and $\mathbf{v}$ are linearly dependent.

1.11 Least-Squares Problems

1.11.1 Least-Squares Solutions

Definition 1.11.1: Least-Squares Solution

Given an $m \times n$ matrix A and $\mathbf{b} \in \mathbb{R}^m$, a *least-squares solution* of $A\mathbf{x} = \mathbf{b}$ is a vector $\hat{\mathbf{x}} \in \mathbb{R}^n$ such that

$$\|\mathbf{b} - A\hat{\mathbf{x}}\| \leq \|\mathbf{b} - A\mathbf{x}\| \quad \text{for all } \mathbf{x} \in \mathbb{R}^n$$

The quantity $\|\mathbf{b} - A\hat{\mathbf{x}}\|$ is called the *least-squares error*.

Remark.

> Since $A\mathbf{x}$ ranges over $\operatorname{Col} A$ as $\mathbf{x}$ varies in $\mathbb{R}^n$, the least-squares solution minimizes the distance from $\mathbf{b}$ to $\operatorname{Col} A$. By the Best Approximation Theorem, $A\hat{\mathbf{x}} = \operatorname{proj}_{\operatorname{Col} A} \mathbf{b}$.

Theorem 1.11.2: Normal Equations

The set of least-squares solutions of $A\mathbf{x} = \mathbf{b}$ coincides with the (nonempty) solution set of the *normal equations*:

$$A^T A \mathbf{x} = A^T \mathbf{b}$$

Proof

Since $A\hat{\mathbf{x}} = \operatorname{proj}_{\operatorname{Col} A} \mathbf{b}$, we have $\mathbf{b} - A\hat{\mathbf{x}} \in (\operatorname{Col} A)^\perp = \operatorname{Nul} A^T$. Thus $A^T(\mathbf{b} - A\hat{\mathbf{x}}) = \mathbf{0}$, giving $A^T A\hat{\mathbf{x}} = A^T\mathbf{b}$. □

Algorithm for Least-Squares Solutions.

1. Compute $A^T A$ and $A^T\mathbf{b}$.
2. Row reduce $[A^T A \mid A^T\mathbf{b}]$ to find all solutions $\hat{\mathbf{x}}$.

Example 1.11.1

Find the least-squares solution of $A\mathbf{x} = \mathbf{b}$ where $A = \begin{bmatrix} 4 & 0 \\ 0 & 2 \\ 1 & 1 \end{bmatrix}$, $\mathbf{b} = \begin{bmatrix} 2 \\ 0 \\ 11 \end{bmatrix}$.

$A^T A = \begin{bmatrix} 17 & 1 \\ 1 & 5 \end{bmatrix}$, $A^T\mathbf{b} = \begin{bmatrix} 19 \\ 11 \end{bmatrix}$. Solving $A^T A\hat{\mathbf{x}} = A^T\mathbf{b}$ gives $\hat{\mathbf{x}} = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$.

Theorem 1.11.3: Unique Least-Squares Solution

The following are equivalent for an $m \times n$ matrix A:

(a) The equation $A\mathbf{x} = \mathbf{b}$ has a unique least-squares solution for each $\mathbf{b} \in \mathbb{R}^m$.

(b) The columns of A are linearly independent.

(c) The matrix $A^T A$ is invertible.

When these hold, $\hat{\mathbf{x}} = (A^T A)^{-1} A^T \mathbf{b}$.

Theorem 1.11.4: Least Squares via QR Factorization

Let $A = QR$ be a QR factorization of A with linearly independent columns. Then for each $\mathbf{b} \in \mathbb{R}^m$, the unique least-squares solution of $A\mathbf{x} = \mathbf{b}$ is

$$\hat{\mathbf{x}} = R^{-1} Q^T \mathbf{b}$$

which in practice is found by solving the upper triangular system $R\mathbf{x} = Q^T \mathbf{b}$.

1.11.2 Least-Squares Lines and Linear Models

The Linear Regression Model. Given data points $(x_1, y_1), \ldots, (x_n, y_n)$, we seek the *least-squares line* $y = \beta_0 + \beta_1 x$ that minimizes the sum of squared residuals. Write this as the inconsistent system $X\beta = \mathbf{y}$ where

$$X = \begin{bmatrix} 1 & x_1 \\ 1 & x_2 \\ \vdots & \vdots \\ 1 & x_n \end{bmatrix}, \quad \beta = \begin{bmatrix} \beta_0 \\ \beta_1 \end{bmatrix}, \quad \mathbf{y} = \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{bmatrix}$$

The least-squares solution $\hat{\beta}$ satisfies the normal equations $X^T X \hat{\beta} = X^T \mathbf{y}$.

Example 1.11.2

Find the least-squares line through $(2, 1), (5, 2), (7, 3), (8, 3)$.
$X^T X = \begin{bmatrix} 4 & 22 \\ 22 & 142 \end{bmatrix}$, $X^T \mathbf{y} = \begin{bmatrix} 9 \\ 57 \end{bmatrix}$.
Solving gives $\hat{\beta}_0 = 2/7$ and $\hat{\beta}_1 = 5/14$, so the least-squares line is $y = \frac{2}{7} + \frac{5}{14}x$.

Chapter 2

Multivariable Calculus

2.1 Limits, Continuity, and Derivatives

Definition 2.1.1: Function

A *function* $f : D \to E$ is a rule that assigns to each element $x \in D$ exactly one element $f(x) \in E$. The set D is the *domain*, and the set of all actual output values is the *range*.

Informal Definition of Limit The statement $\lim_{x\to a} f(x) = L$ means that $f(x)$ can be made arbitrarily close to L by taking x sufficiently close to a, from both the left and the right, but with $x \neq a$.

Right- and Left-Hand Limits $\lim_{x\to a^+} f(x) = L$ means that $f(x)$ can be made arbitrarily close to L by taking x sufficiently close to a with $x > a$. Similarly. $\lim_{x\to a^-} f(x) = L$ means that x approaches a from the left, so $x < a$. The two-sided limit exists and equals L if and only if:

$$\lim_{x\to a} f(x) = L \iff \lim_{x\to a^-} f(x) = L \text{ and } \lim_{x\to a^+} f(x) = L$$

Vertical Asymptote The line $x = a$ is a *vertical asymptote* of the graph of f if at least one of the following is true:

$$\lim_{x\to a^-} f(x) = \pm\infty, \qquad \lim_{x\to a^+} f(x) = \pm\infty$$

Theorem 2.1.2: Squeeze Theorem

Suppose $g(x) \le f(x) \le h(x)$ for all x near a. If $\lim_{x\to a} g(x) = L$ and $\lim_{x\to a} h(x) = L$, then $\lim_{x\to a} f(x) = L$.

Definition 2.1.3: Formal Definition of Limit

Let f be defined near a, except possibly at a. We say $\lim_{x\to a} f(x) = L$ if for every $\varepsilon > 0$, there exists $\delta > 0$ such that whenever $0 < |x - a| < \delta$, we have $|f(x) - L| < \varepsilon$.

Example 2.1.1

Proof that $\lim_{x\to 2} \frac{x^2-4}{x-2} = 4$. Let $\varepsilon > 0$ be given. Choose $\delta = \varepsilon$. If $0 < |x-2| < \delta$, then $x \neq 2$, so $\left|\frac{x^2-4}{x-2} - 4\right| = \left|\frac{(x-2)(x+2)}{x-2} - 4\right| = |x+2-4| = |x-2| < \delta = \varepsilon$. Thus, by the ε-δ definition, $\lim_{x\to 2} \frac{x^2-4}{x-2} = 4$.

Continuity at a Point A function $f : D \to \mathbb{R}$ is *continuous at* $a \in D$ if $\lim_{x\to a} f(x) = f(a)$.

Continuity on an Interval A function is *continuous on an interval* if it is continuous at every point of that interval. At endpoints, continuity is understood using the appropriate one-sided limit.

Types of Discontinuity. Typical discontinuities include:

- *removable discontinuity*: the limit exists but $f(a)$ is missing or has the wrong value;
- *jump discontinuity*: the one-sided limits exist but are not equal;
- *infinite discontinuity*: the function tends to $\pm\infty$;
- *oscillatory discontinuity*: the function oscillates and has no limiting value.

Theorem 2.1.4: Intermediate Value Theorem

Suppose f is continuous on the closed interval $[a, b]$. If N is any number between $f(a)$ and $f(b)$, then there exists at least one $c \in [a, b]$ such that $f(c) = N$.

Limit at Infinity $\lim_{x\to+\infty} f(x) = L$ means that $\forall\varepsilon > 0$, there exists $N > 0$ such that if $x > N$, then $|f(x) - L| < \varepsilon$. Similarly, $\lim_{x\to-\infty} f(x) = L$ means that $\forall\varepsilon > 0$, there exists $N > 0$ such that if $x < -N$, then $|f(x) - L| < \varepsilon$.

Horizontal Asymptote The line $y = L$ is a *horizontal asymptote* if

$$\lim_{x\to+\infty} f(x) = L \quad \text{or} \quad \lim_{x\to-\infty} f(x) = L$$

Tangent Line If the limit $m = \lim_{x\to a} \frac{f(x)-f(a)}{x-a}$ exists, then the tangent line to the graph of f at $(a, f(a))$ is

$$y - f(a) = m(x - a)$$

Definition 2.1.5: Derivative at a Point

The derivative of f at a is

$$f'(a) = \lim_{x\to a} \frac{f(x) - f(a)}{x - a} = \lim_{h\to 0} \frac{f(a+h) - f(a)}{h}$$

The derivative is the slope of the tangent line and the instantaneous rate of change of f at a.

Derivative Function The derivative of f is the function defined at all x where this limit exists:

$$f'(x) = \lim_{h\to 0} \frac{f(x+h) - f(x)}{h}$$

Differentiability Implies Continuity diffimpliescont If f is differentiable at a, then f is continuous at a.

The converse is false. A function can be continuous but not differentiable. $f(x) = |x|$.

Basic Differentiation Rules Let $c \in \mathbb{R}$ and suppose f, g are differentiable. Then

$$\frac{d}{dx}[c] = 0 \qquad \frac{d}{dx}[x^n] = nx^{n-1} \qquad \frac{d}{dx}[cf(x)] = cf'(x) \qquad \frac{d}{dx}[f(x) \pm g(x)] = f'(x) \pm g'(x)$$

Product Rule and Quotient Rule productrule If f and g are differentiable, then fg is differentiable and

$$\frac{d}{dx}[f(x)g(x)] = f'(x)g(x) + f(x)g'(x) \qquad \frac{d}{dx}\left[\frac{f(x)}{g(x)}\right] = \frac{f'(x)g(x) - f(x)g'(x)}{[g(x)]^2}$$

Derivative Formulas for Trigonometric Functions

$$\frac{d}{dx}\sin x = \cos x \qquad \frac{d}{dx}\cos x = -\sin x \qquad \frac{d}{dx}\tan x = \sec^2 x$$

$$\frac{d}{dx}\cot x = -\csc^2 x \qquad \frac{d}{dx}\sec x = \sec x \tan x \qquad \frac{d}{dx}\csc x = -\csc x \cot x$$

Theorem 2.1.6: Chain Rule

If g is differentiable at x and f is differentiable at $g(x)$, let $y = f(u)$ and $u = g(x)$, then

$$\frac{d}{dx}f(g(x)) = f'(g(x))g'(x) \qquad \frac{dy}{dx} = \frac{dy}{du}\frac{du}{dx}$$

Proof

Assume first that $g(x) \neq g(a)$ for x sufficiently close to a, $x \neq a$. Then

$$(f \circ g)'(a) = \lim_{x \to a} \frac{f(g(x)) - f(g(a))}{x - a} = \lim_{x \to a} \frac{f(g(x)) - f(g(a))}{g(x) - g(a)} \cdot \frac{g(x) - g(a)}{x - a} = f'(g(a))g'(a)$$

□

Inverse Function Derivative inversefunctionderivative

$$(f^{-1})'(x) = \frac{1}{f'(f^{-1}(x))}$$

Exponential and Logarithm Derivatives

$$\frac{d}{dx}e^x = e^x \qquad \frac{d}{dx}a^x = a^x \ln a$$

$$\frac{d}{dx}\ln x = \frac{1}{x} \qquad \frac{d}{dx}\log_a x = \frac{1}{x \ln a}$$

Inverse Trigonometric Derivatives

$$\frac{d}{dx}\arcsin x = \frac{1}{\sqrt{1-x^2}} \qquad \frac{d}{dx}\arccos x = -\frac{1}{\sqrt{1-x^2}} \qquad \frac{d}{dx}\arctan x = \frac{1}{1+x^2}$$

Implicit Derivative A function may be defined implicitly by a relation between x and y; instead of solving for y explicitly, differentiate both sides with respect to x, treating y as a function of x, and apply the chain rule when differentiating expressions involving y, e.g. $\frac{d}{dx}[y] = y'$, $\frac{d}{dx}[y^2] = 2yy'$, $\frac{d}{dx}[\sin y] = \cos(y)y'$.

Linearization If f is differentiable at $x = a$, the **linearization** of f at a is $L(x) = f(a) + f'(a)(x - a)$.

Differential If $y = f(x)$, then $dy = f'(x)\,dx$. The actual change is $\Delta y = f(x+\Delta x) - f(x) \approx dy = f'(x)\Delta x, \Delta x \to 0$.

Theorem 2.1.7: Extreme Value Theorem

If f is continuous on a closed interval $[a, b]$, then f attains both an absolute maximum and minimum on $[a, b]$.

Local and Absolute Extremum

- f has a **local maximum** at c if $f(c) \geq f(x)$ for all x sufficiently close to c.
- f has a **local minimum** at c if $f(c) \leq f(x)$ for all x sufficiently close to c.
- f has an **absolute maximum** at c on a domain D if $f(c) \geq f(x) \quad$ for all $x \in D$.
- f has an **absolute minimum** at c on a domain D if $f(c) \leq f(x) \quad$ for all $x \in D$.

Critical Number A number c in the domain of f is a **critical number** if $f'(c) = 0$ or $f'(c)$ does not exist.

Closed Interval Method To find absolute extrema of a continuous function f on $[a, b]$:

1. Find all critical numbers in (a, b).
2. Evaluate f at the critical numbers. Evaluate $f(a)$ and $f(b)$. Compare all values.

Theorem 2.1.8: Rolle's Theorem

Suppose f is continuous on $[a, b]$, f is differentiable on (a, b), and $f(a) = f(b)$, then $\exists c \in (a, b)$ such that

$$f'(c) = 0$$

Proof

Since f is continuous on $[a, b]$, by the Extreme Value Theorem, f attains a maximum and a minimum on $[a, b]$. If f is constant on $[a, b]$, then $f'(x) = 0$ for every $x \in (a, b)$, so the result holds. Assume f is not constant. Since $f(a) = f(b)$, at least one of the maximum or minimum values of f is attained at some interior point $c \in (a, b)$. Thus f has a local maximum or local minimum at c. Because f is differentiable on (a, b), Fermat's Theorem gives $f'(c) = 0$. Therefore, there exists $c \in (a, b)$ such that $f'(c) = 0$. $\square$

Theorem 2.1.9: Mean Value Theorem

Suppose f is continuous on $[a, b]$ and f is differentiable on (a, b) then there exists $c \in (a, b)$ such that

$$f'(c) = \frac{f(b) - f(a)}{b - a}$$

Proof

Let $m = \frac{f(b)-f(a)}{b-a}$. Consider the secant line $L(x) = f(a) + m(x - a)$, and define $g(x) = f(x) - L(x)$. Since f is continuous on $[a, b]$ and differentiable on (a, b), and L is linear, g is also continuous on $[a, b]$ and differentiable on (a, b). Moreover, $g(a) = f(a) - L(a) = f(a) - f(a) = 0$, and $g(b) = f(b) - L(b) = f(b) - (f(a) + m(b - a)) =$

$f(b) - (f(a) + f(b) - f(a)) = 0$. Thus $g(a) = g(b)$. By Rolle's Theorem, there exists $c \in (a, b)$ such that $g'(c) = 0$. But $g'(x) = f'(x) - L'(x) = f'(x) - m$. Therefore, $f'(c) = m = \frac{f(b)-f(a)}{b-a}$. □

First Derivative Test Let c be a critical number:

1. If f' changes from positive to negative at c, then f has a local maximum at c.
2. If f' changes from negative to positive at c, then f has a local minimum at c.
3. If f' does not change sign at c, then f has no local extremum at c.

Concavity

1. f is **concave up** on an interval if $f''(x) > 0$ on that interval.
2. f is **concave down** on an interval if $f''(x) < 0$ on that interval.

Second Derivative Test Suppose $f'(c) = 0$:

1. If $f''(c) > 0$, then f has a local minimum at c.
2. If $f''(c) < 0$, then f has a local maximum at c.
3. If $f''(c) = 0$, the test is inconclusive.

Inflection Point A point $x = c$ is an **inflection point** if f is continuous at c and concavity changes at c. Possible candidates occur when $f''(c) = 0$ or $f''(c)$ does not exist

Theorem 2.1.10: L'Hopital's Rule

If $\lim_{x\to a} f(x) = \lim_{x\to a} g(x) = 0$ or both limits are infinite, and if $\lim_{x\to a} \frac{f'(x)}{g'(x)}$ exists or equals $\pm\infty$, then

$$\lim_{x\to a} \frac{f(x)}{g(x)} = \lim_{x\to a} \frac{f'(x)}{g'(x)}$$

2.2 Integrations and the Fundamental Theorem of Calculus

Area Approximation On $[a, b]$, $\Delta x = \frac{b-a}{n}$. Right endpoint: $x_i = a + i\Delta x$. Left endpoint: $x_{i-1} = a + (i-1)\Delta x$. Midpoint: $\bar{x}_i = a + \left(i - \frac{1}{2}\right)\Delta x$. The approximate area is $\sum_{i=1}^{n} f(x_i^*)\Delta x$.

Definition 2.2.1: Definite Integral

If the limit exists, let $\Delta x = \frac{b-a}{n}$:

$$\int_a^b f(x)\,dx = \lim_{n\to\infty} \sum_{i=1}^{n} f(x_i^*)\Delta x$$

A definite integral is signed area. Area above the x-axis is positive; area below is negative.

Properties

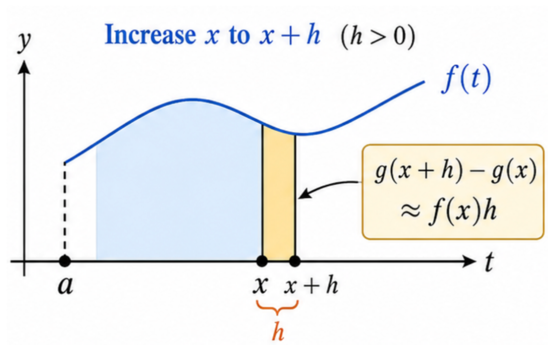
$$\int_a^a f(x)\,dx = 0 \qquad \int_a^b f(x)\,dx = -\int_b^a f(x)\,dx \qquad \int_a^b cf(x)\,dx = c\int_a^b f(x)\,dx$$

$$\int_a^b [f(x)+g(x)]\,dx = \int_a^b f(x)\,dx + \int_a^b g(x)\,dx \qquad \int_a^b f(x)\,dx = \int_a^c f(x)\,dx + \int_c^b f(x)\,dx$$

Antiderivative F is an antiderivative of f if $F'(x) = f(x)$. All antiderivatives are $F(x) + C$.

Theorem 2.2.2: Fundamental Theorem of Calculus Part 1

If f is continuous and $g(x) = \int_a^x f(t)\,dt$ then $g'(x) = f(x)$.



The rate of change of accumulated area is the current height of the function.

Proof

By the definition of derivative, $g'(x) = \lim_{h\to 0} \frac{g(x+h)-g(x)}{h}$. Since $g(x+h) - g(x) = \int_a^{x+h} f(t)\,dt - \int_a^x f(t)\,dt = \int_x^{x+h} f(t)\,dt$, we have $g'(x) = \lim_{h\to 0} \frac{1}{h}\int_x^{x+h} f(t)\,dt$. Because f is continuous, the average value of f on $[x, x+h]$ approaches $f(x)$ as $h \to 0$. Hence $\lim_{h\to 0} \frac{1}{h}\int_x^{x+h} f(t)\,dt = f(x)$. Therefore, $g'(x) = f(x)$. $\square$

Theorem 2.2.3: Fundamental Theorem of Calculus Part 2

If f is continuous on $[a, b]$ and $F'(x) = f(x)$, then

$$\int_a^b f(x)\,dx = F(b) - F(a)$$

Adding up infinitely many tiny changes $f(x)dx$ gives the total change $F(b) - F(a)$ from a to b.

Proof

Define $g(x) = \int_a^x f(t)\,dt$. By the Fundamental Theorem of Calculus Part 1, $g'(x) = f(x)$. Since $F'(x) = f(x)$, we have $g'(x) = F'(x)$. Therefore, for some constant C, $g(x) = F(x) + C$. Evaluating at $x = a$, we get $g(a) = \int_a^a f(t)\,dt = 0$. Hence $C = -F(a)$. Therefore, $g(x) = F(x) - F(a)$. Setting $x = b$, we obtain the result. $\square$

Substitution Rule If $u = g(x), du = g'(x)\,dx$, then

$$\int f(g(x))g'(x)\,dx = \int f(u)\,du$$

For definite integrals, change the bounds:

$$\int_a^b f(g(x))g'(x)\,dx = \int_{g(a)}^{g(b)} f(u)\,du$$

Integration by Parts

$$\int u\,dv = uv - \int v\,du$$

For definite integrals,

$$\int_a^b u\,dv = [uv]_a^b - \int_a^b v\,du$$

Partial Fractions For a proper rational function $P(x)/Q(x)$ (where $\deg P < \deg Q$):

- **Distinct linear:** $(x-a)(x-b) \to \frac{A}{x-a} + \frac{B}{x-b}$
- **Repeated linear:** $(x-a)^r \to \frac{A_1}{x-a} + \cdots + \frac{A_r}{(x-a)^r}$
- **Irreducible quadratic:** $ax^2+bx+c \to \frac{Ax+B}{ax^2+bx+c}$

Key integrals:

$$\int \frac{dx}{ax+b} = \frac{1}{a}\ln|ax+b| + C, \qquad \int \frac{dx}{x^2+a^2} = \frac{1}{a}\arctan\left(\frac{x}{a}\right) + C$$

Infinite Intervals

$$\int_a^\infty f(x)\,dx = \lim_{t\to\infty}\int_a^t f(x)\,dx$$

$$\int_{-\infty}^b f(x)\,dx = \lim_{t\to-\infty}\int_t^b f(x)\,dx$$

For $\int_{-\infty}^{\infty} f(x)\,dx$ split at any c:

$$\int_{-\infty}^{\infty} f(x)\,dx = \int_{-\infty}^{c} f(x)\,dx + \int_c^\infty f(x)\,dx.$$

Comparison Theorem If $0 \le f(x) \le g(x)$ on $[a,\infty)$, then:

1. If $\int_a^\infty g(x)\,dx$ converges, then $\int_a^\infty f(x)\,dx$ converges.
2. If $\int_a^\infty f(x)\,dx$ diverges, then $\int_a^\infty g(x)\,dx$ diverges.

Absolute Area If f crosses the x-axis, geometric area is below, split at zeros of f and remove negative signed area:

$$A = \int_a^b |f(x)|\,dx$$

Area Between Curves If $f(x) \ge g(x)$ on $[a,b]$, then

$$A = \int_a^b [f(x) - g(x)]\,dx$$

Surface Area If $y = f(x) \ge 0$ and f' is continuous on $[a,b]$, then rotating the curve around the x-axis gives

$$S = 2\pi \int_a^b f(x)\sqrt{1+[f'(x)]^2}\,dx$$

2.3 Sequences, Series, and Power Series

Limit of a sequence A sequence (a_n) converges to $L \in \mathbb{R}$ if for every $\varepsilon > 0$, there exists $N \in \mathbb{N}$ such that

$$n \geq N \quad \Longrightarrow \quad |a_n - L| < \varepsilon \Longrightarrow \lim_{n\to\infty} a_n = L$$

Convergence of a series The series $\sum_{n=1}^{\infty} a_n$ converges to S if $\lim_{N\to\infty} s_N = S$.

- Geometric series, converges if $|r| < 1$:$\sum_{n=0}^{\infty} ar^n = \frac{a}{1-r}$.
- Harmonic series, diverges: $\sum_{n=1}^{\infty} \frac{1}{n}$.

A necessary condition for convergence is $\sum a_n$ *converges* $\Longrightarrow \lim_{n\to\infty} a_n = 0$. *The converse is false.*

Integral Test Let f be continuous, positive, and decreasing on $[1, \infty)$, and let $a_n = f(n)$. Then $\sum_{n=1}^{\infty} a_n$ converges if and only if $\int_1^{\infty} f(x)\,\mathrm{d}x$ converges.

Direct Comparison Test Suppose $0 \leq a_n \leq b_n$ for all sufficiently large n.

- If $\sum b_n$ converges, then $\sum a_n$ converges.
- If $\sum a_n$ diverges, then $\sum b_n$ diverges.

Limit Comparison Test Let $a_n, b_n > 0$ and suppose $\lim_{n\to\infty} \frac{a_n}{b_n} = c$ where $0 < c < \infty$, $\sum a_n, \sum b_n$ either both converge or both diverge.

Alternating Series An alternating series has the form

$$\sum_{n=1}^{\infty} (-1)^{n-1} b_n, \qquad b_n \geq 0$$

Alternating Series Test If $b_{n+1} \leq b_n$ for all sufficiently large n and $\lim_{n\to\infty} b_n = 0$, then $\sum_{n=1}^{\infty} (-1)^{n-1} b_n$ converges.

Absolute and conditional convergence The series $\sum a_n$ converges absolutely if $\sum |a_n|$ converges. It converges conditionally if $\sum a_n$ converges but $\sum |a_n|$ diverges. Absolute convergence is stronger than ordinary convergence:

$$\sum |a_n| \text{ converges } \Longrightarrow \sum a_n \text{ converges}$$

Ratio test For a series $\sum a_n$, suppose $L = \lim_{n\to\infty} \left|\frac{a_{n+1}}{a_n}\right|$, then:

$$L < 1 \Rightarrow \text{absolute convergence}, \qquad L > 1 \Rightarrow \text{divergence}, \qquad L = 1 \Rightarrow \text{inconclusive}$$

Root test For a series $\sum a_n$, suppose $L = \lim_{n\to\infty} \sqrt[n]{|a_n|}$, then

$$L < 1 \Rightarrow \text{absolute convergence}, \qquad L > 1 \Rightarrow \text{divergence}, \qquad L = 1 \Rightarrow \text{inconclusive}$$

Power Series A power series centered at a has the form

$$\sum_{n=0}^{\infty} c_n (x-a)^n$$

There exists a radius of convergence $R \in [0, \infty]$ such that the series converges for $|x - a| < R$ and diverges for $|x - a| > R$. Endpoints must be checked separately. The interval of convergence is obtained by:

1. applying the ratio or root test to find R
2. checking $x = a - R$ and $x = a + R$ individually

Definition 2.3.1: Taylor and Maclaurin Series

If f is infinitely differentiable near a, its Taylor series at a is below. $a = 0$ gives a Maclaurin series.

$$f(x) = \sum_{n=0}^{\infty} \frac{f^{(n)}(a)}{n!}(x-a)^n$$

Proof

The idea of Taylor expansion is to approximate $f(x)$ near a by a polynomial $P_k(x) = c_0 + c_1(x-a) + c_2(x-a)^2 + \cdots + c_k(x-a)^k$. We choose the coefficients so that P_k has the same derivatives as f at a: $P_k^{(j)}(a) = f^{(j)}(a), j = 0, 1, \ldots, k$. Since $P_k^{(j)}(a) = j!c_j$, we get $c_j = \frac{f^{(j)}(a)}{j!}$. Therefore the k-th Taylor polynomial of f at a is $P_k(x) = \sum_{j=0}^{k} \frac{f^{(j)}(a)}{j!}(x-a)^j$. $\square$

The most important standard Maclaurin series are:

$$e^x = \sum_{n=0}^{\infty} \frac{x^n}{n!}, \qquad \sin x = \sum_{n=0}^{\infty} (-1)^n \frac{x^{2n+1}}{(2n+1)!}$$

$$\cos x = \sum_{n=0}^{\infty} (-1)^n \frac{x^{2n}}{(2n)!}, \qquad \frac{1}{1-x} = \sum_{n=0}^{\infty} x^n, \quad |x| < 1$$

Theorem 2.3.2: Taylor Remainder Theorem

Suppose f is $k+1$ times differentiable on an interval containing a and x, there exists some δ between a and x such that

$$f(x) = \sum_{j=0}^{k} \frac{f^{(j)}(a)}{j!}(x-a)^j + \frac{f^{(k+1)}(\delta)}{(k+1)!}(x-a)^{k+1}$$

Proof

Assume $a < x$. The case $x < a$ is similar. Define R by

$$f(x) = \sum_{j=0}^{k} \frac{f^{(j)}(a)}{j!}(x-a)^j + \frac{R}{(k+1)!}(x-a)^{k+1}$$

We need to show that $R = f^{(k+1)}(\delta)$ for some $\delta \in (a, x)$. Define, for $z \in [a, x]$,

$$g(z) = \sum_{j=0}^{k} \frac{f^{(j)}(z)}{j!}(x-z)^j + \frac{R}{(k+1)!}(x-z)^{k+1}$$

Then $g(a) = f(x), g(x) = f(x)$. Hence by Rolle's theorem, there exists $\delta \in (a, x)$ such that $g'(\delta) = 0$.

$$\begin{aligned} g'(z) &= \sum_{j=0}^{k} \frac{f^{(j+1)}(z)}{j!}(x-z)^j - \sum_{j=1}^{k} \frac{f^{(j)}(z)}{(j-1)!}(x-z)^{j-1} - \frac{R}{k!}(x-z)^k \\ &= \frac{f^{(k+1)}(z)}{k!}(x-z)^k - \frac{R}{k!}(x-z)^k \\ &= \frac{(x-z)^k}{k!}\left(f^{(k+1)}(z) - R\right) \end{aligned}$$

Since $g'(\delta) = 0$, we have $\frac{(x-\delta)^k}{k!}\left(f^{(k+1)}(\delta) - R\right) = 0$. Because $\delta \in (a, x)$, $x - \delta \neq 0$. Therefore $R = f^{(k+1)}(\delta)$.

□

2.4 Functions of Several Variables

3D Coordinate Systems A point in three-dimensional space is written as (x, y, z). The distance between two points $P_1 = (x_1, y_1, z_1)$ and $P_2 = (x_2, y_2, z_2)$ is:

$$d(P_1, P_2) = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2 + (z_2 - z_1)^2}.$$

Vectors A vector is a quantity with magnitude and direction:

$$\mathbf{v} = \langle v_1, v_2, v_3 \rangle \qquad \|\mathbf{v}\| = \sqrt{v_1^2 + v_2^2 + v_3^2}$$

Dot Product The dot product of $\mathbf{a} = \langle a_1, a_2, a_3 \rangle$ and $\mathbf{b} = \langle b_1, b_2, b_3 \rangle$ is

$$\mathbf{a} \cdot \mathbf{b} = a_1 b_1 + a_2 b_2 + a_3 b_3 = \|\mathbf{a}\| \, \|\mathbf{b}\| \cos\theta$$

Projections The projection of $\mathbf{a}$ onto $\mathbf{b}$ is

$$\operatorname{proj}_{\mathbf{b}} \mathbf{a} = \frac{\mathbf{a} \cdot \mathbf{b}}{\|\mathbf{b}\|^2}\mathbf{b}$$

Cross Product The cross product is below which is perpendicular to both $\mathbf{a}$ and $\mathbf{b}$

$$\mathbf{a} \times \mathbf{b} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \end{vmatrix} a$$

$$\|\mathbf{a} \times \mathbf{b}\| = \|\mathbf{a}\| \, \|\mathbf{b}\| \sin\theta$$

Line A line through a point $\mathbf{r}_0 = \langle x_0, y_0, z_0 \rangle$ in direction $\mathbf{v} = \langle a, b, c \rangle$ is

$$\mathbf{r}(t) = \mathbf{r}_0 + t\mathbf{v}$$

$$x = x_0 + at, \qquad y = y_0 + bt, \qquad z = z_0 + ct$$

Plane A plane through (x_0, y_0, z_0) with normal vector $\mathbf{n} = \langle A, B, C \rangle$ is

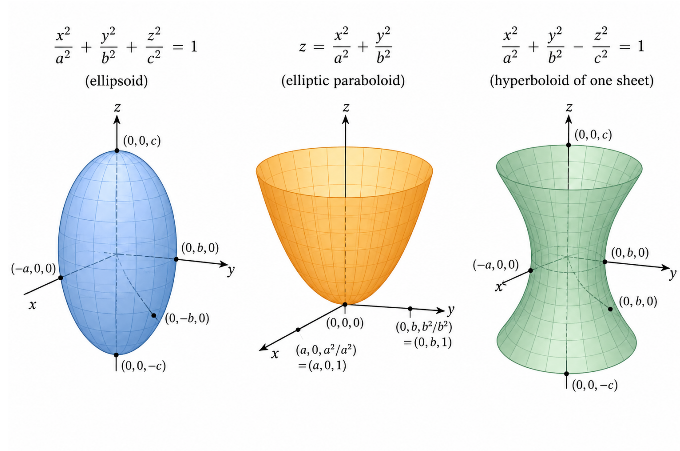
$$A(x - x_0) + B(y - y_0) + C(z - z_0) = 0$$

Quadric Surface Quadric surfaces are second-degree surfaces in x, y, z. Common models include:

$$\frac{x^2}{a^2}+\frac{y^2}{b^2}+\frac{z^2}{c^2}=1 \quad \text{(ellipsoid)}$$

$$z=\frac{x^2}{a^2}+\frac{y^2}{b^2} \quad \text{(elliptic paraboloid)}$$

$$\frac{x^2}{a^2}+\frac{y^2}{b^2}-\frac{z^2}{c^2}=1 \quad \text{(hyperboloid of one sheet)}$$



Vector Functions A vector-valued function has the following form, which describes a curve in space.

$$\mathbf{r}(t)=\langle f(t), g(t), h(t)\rangle$$

The derivative is componentwise, which is tangent to the curve.

$$\mathbf{r}'(t)=\langle f'(t), g'(t), h'(t)\rangle$$

$$\frac{\mathrm{d}}{\mathrm{d}t}(\mathbf{u}\cdot\mathbf{v})=\mathbf{u}'\cdot\mathbf{v}+\mathbf{u}\cdot\mathbf{v}' \qquad \frac{\mathrm{d}}{\mathrm{d}t}(\mathbf{u}\times\mathbf{v})=\mathbf{u}'\times\mathbf{v}+\mathbf{u}\times\mathbf{v}'$$

Arc Length The length of a smooth curve $\mathbf{r}(t)$ for $a \le t \le b$ is

$$L=\int_a^b \|\mathbf{r}'(t)\|\,\mathrm{d}t$$

Curvature Curvature measures how fast the tangent direction changes with respect to arc length:

$$\kappa=\left\|\frac{\mathrm{d}\mathbf{T}}{\mathrm{d}s}\right\| \qquad \mathbf{T}(t)=\frac{\mathbf{r}'(t)}{\|\mathbf{r}'(t)\|}$$

A practical formula is

$$\kappa(t)=\frac{\|\mathbf{r}'(t)\times\mathbf{r}''(t)\|}{\|\mathbf{r}'(t)\|^3}$$

For a plane curve $y=f(x)$,

$$\kappa(x)=\frac{|f''(x)|}{(1+[f'(x)]^2)^{3/2}}$$

Functions of Several Variables A function maps points in a domain to real numbers: $f(x, y)$ or $f(x, y, z)$. Its graph is the surface $z=f(x, y)$. A level curve $f(x, y)=c$ shows where the function has a constant value. A limit $\lim_{(x,y)\to(a,b)} f(x, y)=L$ exists if f approaches L along every path to (a, b). A function is continuous at (a, b) if $\lim_{(x,y)\to(a,b)} f(x, y)=f(a, b)$.

Partial Derivatives These measure rates of change in the x or y direction:

$$f_x(x, y)=\lim_{h\to 0}\frac{f(x+h, y)-f(x, y)}{h} \qquad f_y(x, y)=\lim_{h\to 0}\frac{f(x, y+h)-f(x, y)}{h}$$

Linear Approximation For $z = f(x, y)$, the tangent plane is the linear approximation:

$$z = f(a, b) + f_x(a, b)(x - a) + f_y(a, b)(y - b)$$

$$f(x, y) \approx f(a, b) + f_x(a, b)(x - a) + f_y(a, b)(y - b)$$

Chain Rule For a composition $z = f(x(t), y(t))$, the chain rule is

$$\frac{\mathrm{d}z}{\mathrm{d}t} = f_x \frac{\mathrm{d}x}{\mathrm{d}t} + f_y \frac{\mathrm{d}y}{\mathrm{d}t}$$

For $z = f(x(s, t), y(s, t))$,

$$\frac{\partial z}{\partial s} = f_x \frac{\partial x}{\partial s} + f_y \frac{\partial y}{\partial s} \qquad \frac{\partial z}{\partial t} = f_x \frac{\partial x}{\partial t} + f_y \frac{\partial y}{\partial t}$$

Definition 2.4.1: Directional Derivatives

Let $f : \mathbb{R}^n \to \mathbb{R}$, $\mathbf{a} \in \mathbb{R}^n$, $\mathbf{u} \in \mathbb{R}^n$, $\|\mathbf{u}\| = 1$. The directional derivative of f at $\mathbf{a}$ in the direction $\mathbf{u}$ is defined by

$$D_{\mathbf{u}} f(\mathbf{a}) = \lim_{h \to 0} \frac{f(\mathbf{a} + h\mathbf{u}) - f(\mathbf{a})}{h}$$

Definition 2.4.2: Gradient

Let $f : \mathbb{R}^n \to \mathbb{R}$ be differentiable at a point $\mathbf{a} = (a_1, \ldots, a_n) \in \mathbb{R}^n$. The gradient of f at $\mathbf{a}$ is:

$$\nabla f(\mathbf{a}) = \left\langle \frac{\partial f}{\partial x_1}(\mathbf{a}), \frac{\partial f}{\partial x_2}(\mathbf{a}), \ldots, \frac{\partial f}{\partial x_n}(\mathbf{a}) \right\rangle$$

The gradient of f at $\mathbf{a}$ is the vector of partial derivatives, and for any unit vector $\mathbf{u}$ the directional derivative in the direction of $\mathbf{u}$ is given by the dot product of the gradient with $\mathbf{u}$:

$$D_{\mathbf{u}} f(a, b) = \nabla f(a, b) \cdot \mathbf{u}$$

Proof

Let $\mathbf{u} = \langle u_1, u_2 \rangle$ be a unit vector. Since f is differentiable at (a, b), for $(\Delta x, \Delta y) \to (0, 0)$,

$$f(a + \Delta x, b + \Delta y) = f(a, b) + f_x(a, b)\Delta x + f_y(a, b)\Delta y + o\Big(\sqrt{(\Delta x)^2 + (\Delta y)^2}\Big)$$

Take $\Delta x = hu_1$ and $\Delta y = hu_2$. Since $\|\mathbf{u}\| = 1$,

$$f(a + hu_1, b + hu_2) = f(a, b) + hf_x(a, b)u_1 + hf_y(a, b)u_2 + o(|h|)$$

Therefore,

$$\begin{aligned} D_{\mathbf{u}} f(a, b) &= \lim_{h \to 0} \frac{f(a + hu_1, b + hu_2) - f(a, b)}{h} \\ &= \lim_{h \to 0} \left(f_x(a, b)u_1 + f_y(a, b)u_2 + \frac{o(|h|)}{h} \right) \\ &= f_x(a, b)u_1 + f_y(a, b)u_2 \\ &= \nabla f(a, b) \cdot \mathbf{u} \end{aligned}$$

$\square$

The gradient points in the direction of **steepest ascent**, and the maximum possible directional derivative is $\|\nabla f(\mathbf{a})\|$. Thus, $\nabla f(\mathbf{a})$ gives both the direction and the rate of fastest increase of f at $\mathbf{a}$.

Proof

Assume f is differentiable at $\mathbf{a}$. Then for any $\|\mathbf{u}\| = 1$, $D_{\mathbf{u}}f(\mathbf{a}) = \nabla f(\mathbf{a}) \cdot \mathbf{u}$. By the Cauchy–Schwarz inequality, $\nabla f(\mathbf{a}) \cdot \mathbf{u} \le \|\nabla f(\mathbf{a})\| \, \|\mathbf{u}\| = \|\nabla f(\mathbf{a})\|$. Equality holds when $\mathbf{u} = \frac{\nabla f(\mathbf{a})}{\|\nabla f(\mathbf{a})\|}$, provided $\nabla f(\mathbf{a}) \neq \mathbf{0}$. $\square$

Definition 2.4.3: Jacobian Matrix

Let $F : \mathbb{R}^n \to \mathbb{R}^m$ be a function. If all first-order partial derivatives of $F_1, \ldots, F_m$ exist at $\mathbf{a} \in \mathbb{R}^n$, then the Jacobian matrix of F at $\mathbf{a}$ is the $m \times n$ matrix

$$J_F(\mathbf{a}) = \begin{bmatrix} \frac{\partial F_1}{\partial x_1}(\mathbf{a}) & \frac{\partial F_1}{\partial x_2}(\mathbf{a}) & \cdots & \frac{\partial F_1}{\partial x_n}(\mathbf{a}) \\ \frac{\partial F_2}{\partial x_1}(\mathbf{a}) & \frac{\partial F_2}{\partial x_2}(\mathbf{a}) & \cdots & \frac{\partial F_2}{\partial x_n}(\mathbf{a}) \\ \vdots & \vdots & \ddots & \vdots \\ \frac{\partial F_m}{\partial x_1}(\mathbf{a}) & \frac{\partial F_m}{\partial x_2}(\mathbf{a}) & \cdots & \frac{\partial F_m}{\partial x_n}(\mathbf{a}) \end{bmatrix} = \left[\frac{\partial F_i}{\partial x_j}(\mathbf{a})\right]_{1\le i\le m,\ 1\le j\le n} \qquad F(\mathbf{x}) = \begin{bmatrix} F_1(\mathbf{x}) \\ F_2(\mathbf{x}) \\ \vdots \\ F_m(\mathbf{x}) \end{bmatrix} \qquad \mathbf{x} = (x_1, \ldots, x_n)$$

Definition 2.4.4: Hessian Matrix

Let $f : \mathbb{R}^n \to \mathbb{R}$ be a scalar-valued function, where $\mathbf{x} = (x_1, \ldots, x_n)$. If all second-order partial derivatives of f exist at $\mathbf{a} \in \mathbb{R}^n$, then the Hessian matrix of f at $\mathbf{a}$ is the $n \times n$ matrix

$$H_f(\mathbf{a}) = \begin{bmatrix} \frac{\partial^2 f}{\partial x_1^2}(\mathbf{a}) & \frac{\partial^2 f}{\partial x_1 \partial x_2}(\mathbf{a}) & \cdots & \frac{\partial^2 f}{\partial x_1 \partial x_n}(\mathbf{a}) \\ \frac{\partial^2 f}{\partial x_2 \partial x_1}(\mathbf{a}) & \frac{\partial^2 f}{\partial x_2^2}(\mathbf{a}) & \cdots & \frac{\partial^2 f}{\partial x_2 \partial x_n}(\mathbf{a}) \\ \vdots & \vdots & \ddots & \vdots \\ \frac{\partial^2 f}{\partial x_n \partial x_1}(\mathbf{a}) & \frac{\partial^2 f}{\partial x_n \partial x_2}(\mathbf{a}) & \cdots & \frac{\partial^2 f}{\partial x_n^2}(\mathbf{a}) \end{bmatrix} = \left[\frac{\partial^2 f}{\partial x_i \partial x_j}(\mathbf{a})\right]_{1\le i,j\le n}$$

Unconstrained Extrema Critical points satisfy $\nabla f = \mathbf{0}$.For a function of two variables, define

$$D = f_{xx} f_{yy} - (f_{xy})^2 = |H_f|$$

Then the second derivative test says:

- if $D > 0$ and $f_{xx} > 0$, there is a local minimum;
- if $D > 0$ and $f_{xx} < 0$, there is a local maximum;
- if $D < 0$, there is a saddle point;
- if $D = 0$, the test is inconclusive.

Lagrange Multipliers To optimize $f(x, y)$ subject to $g(x, y) = c$, solve

$$\nabla f = \lambda \nabla g, \qquad g(x, y) = c$$

For two constraints $g(x, y, z) = c$ and $h(x, y, z) = d$, solve

$$\nabla f = \lambda \nabla g + \mu \nabla h, \qquad g = c, \qquad h = d$$

Evaluate f at all candidate points, especially when the constraint set is closed and bounded.

2.5 Multiple Integrals and Coordinate Systems

Double Integrals For a function $f(x, y)$ over a rectangle $R = [a, b] \times [c, d]$, the double integral is

$$\iint_R f(x, y) \, \mathrm{d}A$$

It gives the signed volume under $z = f(x, y)$ over R. Fubini's theorem allows computation by iterated integrals:

$$\iint_R f(x, y) \, \mathrm{d}A = \int_a^b \int_c^d f(x, y) \, \mathrm{d}y \, \mathrm{d}x = \int_c^d \int_a^b f(x, y) \, \mathrm{d}x \, \mathrm{d}y$$

For a general region D, one writes the bounds according to the shape of the domain. A Type I region has

$$D = \{(x, y) : a \le x \le b, \ g_1(x) \le y \le g_2(x)\} \qquad \iint_D f \, \mathrm{d}A = \int_a^b \int_{g_1(x)}^{g_2(x)} f(x, y) \, \mathrm{d}y \, \mathrm{d}x$$

A Type II region has

$$D = \{(x, y) : c \le y \le d, \ h_1(y) \le x \le h_2(y)\} \qquad \iint_D f \, \mathrm{d}A = \int_c^d \int_{h_1(y)}^{h_2(y)} f(x, y) \, \mathrm{d}x \, \mathrm{d}y$$

Polar coordinates are $x = r \cos\theta, y = r \sin\theta$, with area element $\mathrm{d}A = r \, \mathrm{d}r \, \mathrm{d}\theta$. Thus

$$\iint_D f(x, y) \, \mathrm{d}A = \int_\alpha^\beta \int_{r_1(\theta)}^{r_2(\theta)} f(r \cos\theta, r \sin\theta) r \, \mathrm{d}r \, \mathrm{d}\theta$$

Triple Integrals A triple integral over a solid region E is

$$\iiint_E f(x, y, z) \, \mathrm{d}V.$$

It can represent mass when f is density. Rectangular coordinates use $\mathrm{d}V = \mathrm{d}x \, \mathrm{d}y \, \mathrm{d}z$. Cylindrical coordinates are $x = r \cos\theta, y = r \sin\theta, z = z$ with $\mathrm{d}V = r \, \mathrm{d}z \, \mathrm{d}r \, \mathrm{d}\theta$. Spherical coordinates are $x = \rho \sin\phi \cos\theta, y = \rho \sin\phi \sin\theta, z = \rho \cos\phi$, with volume element $\mathrm{d}V = \rho^2 \sin\phi \, \mathrm{d}\rho \, \mathrm{d}\phi \, \mathrm{d}\theta$.

Vector Field A vector field assigns a vector to each point:

$$\mathbf{F}(x, y) = P(x, y)\mathbf{i} + Q(x, y)\mathbf{j}$$

Line Integral For a scalar function f or a vector field $\mathbf{F}$ along a curve C parametrized by $\mathbf{r}(t)$, $a \le t \le b$,

$$\int_C f \, \mathrm{d}s = \int_a^b f(\mathbf{r}(t)) \, \|\mathbf{r}'(t)\| \, \mathrm{d}t$$

$$\int_C \mathbf{F} \cdot \mathrm{d}\mathbf{r} = \int_a^b \mathbf{F}(\mathbf{r}(t)) \cdot \mathbf{r}'(t) \, \mathrm{d}t$$

The fundamental theorem for line integrals says that if $\mathbf{F} = \nabla f$ is conservative, then:

$$\int_C \mathbf{F} \cdot \mathrm{d}\mathbf{r} = f(\mathbf{r}(b)) - f(\mathbf{r}(a))$$

Theorem 2.5.1: Green's theorem

Let C be a positively oriented, piecewise smooth, simple closed curve enclosing a region D. If P and Q have continuous partial derivatives on an open region containing D, then

$$\oint_C P\,\mathrm{d}x + Q\,\mathrm{d}y = \iint_D \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y}\right) \mathrm{d}A$$

Parametric Surface A parametric surface is described by

$$\mathbf{r}(u,v) = \langle x(u,v), y(u,v), z(u,v)\rangle, \qquad (u,v) \in D$$

For fixed u or fixed v, one obtains parameter curves on the surface. The tangent vectors are $\mathbf{r}_u = \frac{\partial \mathbf{r}}{\partial u}, \mathbf{r}_v = \frac{\partial \mathbf{r}}{\partial v}$. A normal vector is $\mathbf{r}_u \times \mathbf{r}_v$. The tangent plane at (u_0, v_0) is determined by the point $\mathbf{r}(u_0, v_0)$ and the normal vector $\mathbf{r}_u(u_0, v_0) \times \mathbf{r}_v(u_0, v_0)$. The surface area is

$$A(S) = \iint_D \|\mathbf{r}_u \times \mathbf{r}_v\|\,\mathrm{d}A$$

Surface Integral For a scalar function f over a parametric surface S given by $\mathbf{r}(u,v)$

$$\iint_S f\,\mathrm{d}S = \iint_D f(\mathbf{r}(u,v))\,\|\mathbf{r}_u \times \mathbf{r}_v\|\,\mathrm{d}A$$

For a vector field $\mathbf{F}$, the flux across an oriented surface S is

$$\iint_S \mathbf{F} \cdot \mathrm{d}\mathbf{S} = \iint_D \mathbf{F}(\mathbf{r}(u,v)) \cdot (\mathbf{r}_u \times \mathbf{r}_v)\,\mathrm{d}A$$

Theorem 2.5.2: Divergence theorem

Let E be a solid region with positively oriented closed boundary surface $S = \partial E$. Then

$$\iint_S \mathbf{F} \cdot \mathrm{d}\mathbf{S} = \iiint_E \operatorname{div} \mathbf{F}\,\mathrm{d}V.$$

Chapter 3

Ordinary Differential Equations

3.1 Introduction to Differential Equations

3.1.1 Definitions and Terminology

Definition 3.1.1: Differential Equation

A *differential equation* is an equation that contains derivatives of one or more unknown functions with respect to one or more independent variables. The unknown function is usually called the *dependent variable*, and the variable with respect to which differentiation is taken is called the *independent variable*.

Ordinary and Partial Differential Equations A DE is called an *ordinary differential equation* (ODE) if it contains only ordinary derivatives with respect to a single independent variable. It is called a *partial differential equation* (PDE) if it contains partial derivatives of one or more unknown functions of two or more independent variables.

Example 3.1.1

The equations $\frac{dy}{dx} + 5y = e^x$ and $\frac{d^2y}{dx^2} - 6y = 0$ are ordinary differential equations. By contrast, $\frac{\partial u}{\partial x} + \frac{\partial u}{\partial y} = 0$ is a partial differential equation.

Order. The *order* of a differential equation is the order of the highest derivative appearing in the equation. In one dependent variable, the general form of an nth-order ODE is

$$F(x, y, y', \ldots, y^{(n)}) = 0$$

If the equation can be solved for the highest derivative, it can be written in *normal form*:

$$\frac{d^n y}{dx^n} = f(x, y, y', \ldots, y^{(n-1)})$$

Linear Ordinary Differential Equation An nth-order ODE is *linear* if it's linear in the dependent variable and all of its derivatives. Its general form is

$$a_n(x)y^{(n)} + a_{n-1}(x)y^{(n-1)} + \cdots + a_1(x)y' + a_0(x)y = g(x)$$

The coefficients may depend on the independent variable x, but not on y or its derivatives.

Example 3.1.2

The equation $y'' - 2y' + y = e^x$ is linear.
The equations $yy' + x = 0$, $y'' + \sin y = 0$ and $(y')^2 + y = 0$ are nonlinear.

Solution of an ODE A function ϕ, defined on an interval I and possessing the required derivatives on I, is called a *solution* of an nth-order ODE if substituting $y = \phi(x)$ into the equation reduces it to an identity on I.

Interval of Definition The interval on which a solution is defined is called the *interval of definition*, *interval of existence*, *interval of validity*, or *domain of the solution*.

Explicit and Implicit Solutions An *explicit solution* expresses the dependent variable directly as a function of the independent variable: $y = \phi(x)$. An *implicit solution* is a relation $G(x, y) = 0$ that defines at least one differentiable function $y = \phi(x)$ satisfying the differential equation on an interval.

Example 3.1.3

The relation $x^2 + y^2 = 25$ is an implicit solution of $\frac{dy}{dx} = -\frac{x}{y}$. On the interval $(-5, 5)$, it gives two explicit branches: $y = \sqrt{25 - x^2}$ and $y = -\sqrt{25 - x^2}$.

Families, Particular Solutions, and Singular Solutions A solution containing one or more constants is called a *family of solutions*. A solution obtained by assigning values to these constants is called a *particular solution*. A *singular solution* is a solution that can't be obtained from the family of solutions by any choice of the arbitrary constants.

Example 3.1.4

The equation $y' = y$ has the one-parameter family of solutions $y = ce^x$. If the condition $y(0) = 3$ is imposed, then $c = 3$, so the particular solution is $y = 3e^x$.

System of ODEs A *system of ordinary differential equations* consists of two or more equations involving derivatives of two or more unknown functions of a single independent variable.

3.1.2 Initial-Value Problems

Definition 3.1.2: Initial-Value Problem

An nth-order *initial-value problem* consists of an nth-order differential equation together with n side conditions imposed at the same point x_0:

$$\frac{d^n y}{dx^n} = f(x, y, y', \dots, y^{(n-1)})$$

subject to

$$y(x_0) = y_0, \qquad y'(x_0) = y_1, \qquad \dots, \qquad y^{(n-1)}(x_0) = y_{n-1}$$

Theorem 3.1.3: Existence and Uniqueness for a First-Order IVP

Let R be a rectangular region in the xy-plane containing the point (x_0, y_0). If $f(x, y)$ and $\dfrac{\partial f}{\partial y}(x, y)$ are continuous on R, then there exists an interval $I_0 = (x_0 - h, x_0 + h)$ for some $h > 0$ on which the initial-value problem

$$\frac{dy}{dx} = f(x, y), \qquad y(x_0) = y_0$$

has a unique solution.

This theorem is local: it guarantees a unique solution only near x_0, not necessarily on the largest possible interval. It gives sufficient, not necessary, conditions. Roughly, f continuous gives existence, and f_y continuous gives uniqueness.

Example 3.1.5

The initial-value problem $\frac{dy}{dx} = x\sqrt{y}, y(0) = 0$ has more than one solution. For example, $y = 0$ and $y = \frac{x^4}{16}$ both satisfy the differential equation and the initial condition. This does not contradict the existence–uniqueness theorem, because $\frac{\partial}{\partial y}\left(x\sqrt{y}\right) = \frac{x}{2\sqrt{y}}$ is not continuous at $y = 0$.

3.2 First-Order Differential Equations

3.2.1 Solution Curves Without an Explicit Solution

Direction Field Consider a first-order differential equation in normal form $\frac{dy}{dx} = f(x, y)$ At each point (x, y) where f is defined, the value $f(x, y)$ gives the slope of a small line segment, called a *lineal element*. A collection of such lineal elements drawn on a grid in the xy-plane is called a *direction field*.

Definition 3.2.1: Autonomous First-Order Differential Equation

A first-order differential equation is called *autonomous* if the independent variable does not appear explicitly. In normal form, it can be written as

$$\frac{dy}{dx} = f(y)$$

Critical Point and Equilibrium Solution A real number c is called a *critical point*, *equilibrium point*, or *stationary point* of $\frac{dy}{dx} = f(y)$ if $f(c) = 0$. In this case, the constant function $y(x) = c$ is a solution, called an *equilibrium solution*.

Phase Line and Phase Portrait For an autonomous equation $\frac{dy}{dx} = f(y)$ a *phase line* is a vertical line on which the critical points are marked. The sign of $f(y)$ on each interval between critical points determines whether solutions are increasing or decreasing. The resulting diagram is called a *one-dimensional phase portrait*.

Attractors, Repellers, and Semi-Stable Points Let c be a critical point of $\frac{dy}{dx} = f(y)$

- If nearby solutions approach c as $x \to \infty$, then c is an *attractor* or *asymptotically stable equilibrium*.
- If nearby solutions move away from c, then c is a *repeller* or *unstable equilibrium*.
- If solutions approach c from one side and move away from c from the other side, then c is *semi-stable*.

Proposition 3.2.2

If $y(x)$ is a solution of the autonomous equation $\frac{dy}{dx} = f(y)$ then for any constant k, $y_1(x) = y(x-k)$ is also a solution. This is called the *translation property.*

3.2.2 Separable Equations

Definition 3.2.3: Separable Equation

A first-order differential equation of the form

$$\frac{dy}{dx} = g(x)h(y)$$

is called a *separable differential equation.*

Procedure. To solve a separable differential equation:

1. Write the equation in separable form $\frac{dy}{dx} = g(x)h(y)$.
2. Assuming $h(y) \neq 0$, separate the variables $\frac{1}{h(y)}\,dy = g(x)\,dx$.
3. Integrate both sides $\int \frac{1}{h(y)}\,dy = \int g(x)\,dx$.
4. Write the solution implicitly as $H(y) = G(x) + C$.

When solving a separable equation, any value r such that $h(r) = 0$ should be checked separately. The constant function $y = r$ may be a solution, but it may be lost after division by $h(y)$.

3.2.3 Linear Equations

Definition 3.2.4: First-Order Linear Equation

A first-order differential equation of the form $a_1(x)\frac{dy}{dx} + a_0(x)y = g(x)$ is called a *linear differential equation* in the dependent variable y. If $a_1(x) \neq 0$, division by $a_1(x)$ gives the standard form

$$\frac{dy}{dx} + P(x)y = f(x)$$

Procedure. To solve a first-order linear equation:

1. Put the equation in standard form $\frac{dy}{dx} + P(x)y = f(x)$.
2. Compute the integrating factor $\mu(x) = e^{\int P(x)\,dx}$.
3. Multiply both sides by $\mu(x)$ and rewrite the left-hand side as $\frac{d}{dx}\big(\mu(x)y\big)$.
4. Integrate both sides and solve for y.

$$y = e^{-\int P(x)\,dx}\left(\int e^{\int P(x)\,dx} f(x)\,dx + C\right)$$

3.2.4 Exact Differential Equations

Definition 3.2.5: Exact Differential Equation

A first-order differential equation written in differential form

$$M(x, y)\, dx + N(x, y)\, dy = 0$$

is called an *exact equation* if there exists a function $\Phi(x, y)$ such that

$$\frac{\partial \Phi}{\partial x} = M(x, y), \qquad \frac{\partial \Phi}{\partial y} = N(x, y)$$

In that case, $d\Phi = M\, dx + N\, dy$ and the solution is given implicitly by $\Phi(x, y) = C$.

Theorem 3.2.6: Criterion for Exactness

Suppose $M(x, y)$ and $N(x, y)$ have continuous first partial derivatives in a rectangular region R. Then $M(x, y)\, dx+ N(x, y)\, dy$ is an exact differential in R if and only if

$$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$$

Procedure.

1. Integrate M with respect to x: $\Phi(x, y) = \int M(x, y)\, dx + g(y)$.
2. Differentiate this expression with respect to y: $\Phi_y = \frac{\partial}{\partial y} \left(\int M(x, y)\, dx \right) + g'(y)$.
3. Set $\Phi_y = N(x, y)$ and solve for $g'(y)$. Integrate to find $g(y)$.
4. Write the solution as $\Phi(x, y) = C$.

Integrating Factors for Nonexact Equations. If $M(x, y)\, dx + N(x, y)\, dy = 0$ is not exact, it may become exact after multiplication by a nonzero function $\mu(x, y)$: $\mu M\, dx + \mu N\, dy =$. Two useful special cases are:

$$\begin{cases} \mu(x) = \exp\left(\int \frac{M_y - N_x}{N}\, dx \right), & \text{if } \frac{M_y - N_x}{N} \text{ depends only on } x \\ \mu(y) = \exp\left(\int \frac{N_x - M_y}{M}\, dy \right), & \text{if } \frac{N_x - M_y}{M} \text{ depends only on } y \end{cases}$$

3.2.5 Solutions by Substitutions

Definition 3.2.7: Substitution Method

For a first-order equation $\frac{dy}{dx} = f(x, y)$ introduce a substitution $y = g(x, u), u = u(x)$. By the chain rule,

$$\frac{dy}{dx} = \frac{\partial g}{\partial x}(x, u) + \frac{\partial g}{\partial u}(x, u) \frac{du}{dx}$$

After substitution, the original equation may become a simpler equation $\frac{du}{dx} = F(x, u)$. If this new equation is separable or linear, it can be solved by earlier methods. Finally, substitute back to obtain y.

Solving a Homogeneous Equation. If the equation is homogeneous, use the substitution

$$y = ux, \qquad \frac{dy}{dx} = u + x\frac{du}{dx}$$

Solving a Bernoulli Equation. A differential equation of the form $\frac{dy}{dx} + P(x)y = f(x)y^n$ is called a *Bernoulli equation.* For $n \neq 0, 1$, it can be reduced to a linear equation by the substitution

$$u = y^{1-n}, \qquad \frac{du}{dx} = (1-n)y^{-n}\frac{dy}{dx}$$

Reduction to Separation of Variables A differential equation of the form $\frac{dy}{dx} = f(Ax + By + C), B \neq 0$ can be reduced to a separable equation by the substitution

$$u = Ax + By + C, \qquad \frac{du}{dx} = A + B\frac{dy}{dx}$$

3.2.6 A Numerical Method

Definition 3.2.8: Euler's Method

Consider the initial-value problem $y' = f(x, y), y(x_0) = y_0$, Euler's method approximates the solution by repeatedly following tangent lines:

$$x_n = x_0 + nh, \qquad y_{n+1} = y_n + hf(x_n, y_n), \qquad n = 0, 1, 2, \ldots$$

3.3 Higher-Order Differential Equations

3.3.1 Theory of Linear Equations

Linear nth-Order Differential Equation A linear nth-order differential equation has the form

$$a_n(x)y^{(n)} + a_{n-1}(x)y^{(n-1)} + \cdots + a_1(x)y' + a_0(x)y = g(x)$$

where $a_n(x) \neq 0$ on an interval I. If $g(x) = 0$, the equation is *homogeneous*; otherwise it is *nonhomogeneous*. An nth-order initial-value problem specifies the value of y and its first $n-1$ derivatives at the same point x_0:

$$y(x_0) = y_0, \qquad y'(x_0) = y_1, \qquad \ldots, \qquad y^{(n-1)}(x_0) = y_{n-1}$$

Theorem 3.3.1: Existence and Uniqueness for Linear IVPs

Suppose $a_n, a_{n-1}, \ldots, a_0$ and g are continuous on an interval I, and $a_n(x) \neq 0$ for every $x \in I$. Then for any $x_0 \in I$, the corresponding nth-order linear initial-value problem has a unique solution on I.

This theorem is much stronger than the first-order local theorem: for linear equations, the solution exists on the whole interval on which the coefficient functions are continuous and the leading coefficient is nonzero.

Boundary-Value Problems. A boundary-value problem imposes conditions at different points, for example $y(a) = y_0, y(b) = y_1$. Unlike initial-value problems, a boundary-value problem may have infinitely many solutions, exactly one solution, or no solution.

Differential Operator Notation. Let $D = \frac{d}{dx}$. Then $D^n y = \frac{d^n y}{dx^n}$. The linear differential operator is

$$L = a_n(x)D^n + a_{n-1}(x)D^{n-1} + \cdots + a_1(x)D + a_0(x)$$

so the differential equation can be written compactly as $L[y] = g(x)$.

Proposition 3.3.2

The operator L is linear, where α and β are constants:

$$L[\alpha f + \beta g] = \alpha L[f] + \beta L[g],$$

Theorem 3.3.3: Superposition Principle for Homogeneous Equations

If $y_1, \ldots, y_k$ are solutions of the homogeneous equation $L[y] = 0$ on an interval I, then every linear combination $y = c_1 y_1 + \cdots + c_k y_k$ is also a solution on I.

Definition 3.3.4: Linear Independence and the Wronskian

Functions $f_1, \ldots, f_n$ are linearly dependent on I if there exist constants $c_1, \ldots, c_n$, not all zero, such that

$$c_1 f_1(x) + \cdots + c_n f_n(x) = 0$$

for every $x \in I$. Otherwise they are linearly independent. The *Wronskian* of $f_1, \ldots, f_n$ is

$$W(f_1, \ldots, f_n) = \begin{vmatrix} f_1 & f_2 & \cdots & f_n \\ f_1' & f_2' & \cdots & f_n' \\ \vdots & \vdots & & \vdots \\ f_1^{(n-1)} & f_2^{(n-1)} & \cdots & f_n^{(n-1)} \end{vmatrix}$$

Wronskian Criterion for Solutions wronskiancriterion Let $y_1, \ldots, y_n$ be n solutions of a homogeneous linear nth-order differential equation on I. Then they are linearly independent on I if and only if for every $x \in I$:

$$W(y_1, \ldots, y_n)(x) \neq 0$$

Fundamental Set of Solutions A set of n linearly independent solutions of a homogeneous linear nth-order differential equation is called a *fundamental set of solutions*.

Theorem 3.3.5: General Solution of a Homogeneous Linear Equation

If $y_1, \ldots, y_n$ form a fundamental set of solutions for $L[y] = 0$ on I, then the general solution is

$$y = c_1 y_1 + \cdots + c_n y_n$$

Theorem 3.3.6: General Solution of a Nonhomogeneous Linear Equation

If y_p is one particular solution of $L[y] = g(x)$ and $y_c = c_1y_1 + \cdots + c_ny_n$ is the general solution of the associated homogeneous equation $L[y] = 0$, then the general solution of the nonhomogeneous equation is

$$y = y_c + y_p$$

Proposition 3.3.7

If $y_{p1}, \dots, y_{pk}$ are particular solutions of

$$L[y] = g_1(x), \qquad \dots, \qquad L[y] = g_k(x)$$

then $y_p = y_{p1} + \cdots + y_{pk}$ is a particular solution of

$$L[y] = g_1(x) + \cdots + g_k(x)$$

3.3.2 Reduction of Order

Theorem 3.3.8: Reduction of Order Formula

Let $y_1(x)$ be a nonzero solution of

$$y'' + P(x)y' + Q(x)y = 0$$

on an interval I. Then a second solution is

$$y_2(x) = y_1(x) \int \frac{e^{-\int P(x)\,dx}}{[y_1(x)]^2}\,dx$$

Procedure.

1. Put the equation in standard form $y'' + P(x)y' + Q(x)y = 0$.
2. Identify a known nonzero solution y_1.
3. Use $y_2 = y_1 \int \frac{e^{-\int P(x)\,dx}}{y_1^2}\,dx$.
4. Write the general solution as $y = c_1y_1 + c_2y_2$.

3.3.3 Homogeneous Linear Equations with Constant Coefficients

Auxiliary Equation. For $a_ny^{(n)} + a_{n-1}y^{(n-1)} + \cdots + a_1y' + a_0y = 0$, try $y = e^{mx}$. Substitution gives the *auxiliary equation* $a_nm^n + a_{n-1}m^{n-1} + \cdots + a_1m + a_0 = 0$. The roots of this polynomial determine the solution.

Second-Order Case. For $ay'' + by' + cy = 0$ the auxiliary equation is $am^2 + bm + c = 0$.

Root type	General solution
$m_1 \neq m_2$ real	$y = c_1e^{m_1x} + c_2e^{m_2x}$
$m_1 = m_2 = m$	$y = c_1e^{mx} + c_2xe^{mx}$
$m = \alpha \pm i\beta$	$y = e^{\alpha x}\big(c_1\cos\beta x + c_2\sin\beta x\big)$

Higher-Order Roots. If m is a real root of multiplicity k, then the corresponding solutions are

$$e^{mx}, \quad xe^{mx}, \quad x^2e^{mx}, \quad \dots, \quad x^{k-1}e^{mx}$$

If $\alpha \pm i\beta$ is a complex conjugate pair of multiplicity k, then the real solutions are

$$e^{\alpha x}\cos\beta x, \quad xe^{\alpha x}\cos\beta x, \quad \dots, \quad x^{k-1}e^{\alpha x}\cos\beta x$$

$$e^{\alpha x}\sin\beta x, \quad xe^{\alpha x}\sin\beta x, \quad \dots, \quad x^{k-1}e^{\alpha x}\sin\beta x$$

3.3.4 Undetermined Coefficients: Superposition Approach

When the Method Applies. The method of undetermined coefficients applies to nonhomogeneous linear differential equations with constant coefficients, $a_ny^{(n)} + a_{n-1}y^{(n-1)} + \dots + a_1y' + a_0y = g(x)$, when $g(x)$ is a finite sum of constants, polynomials, exponentials, sines, cosines, or products of these functions.

Procedure.

1. Solve the associated homogeneous equation to find y_c.
2. Guess the form of a particular solution y_p based on $g(x)$.
3. Substitute y_p into the original equation and solve for the unknown coefficients.
4. Write the general solution as $y = y_c + y_p$.

Common Trial Forms.

$g(x)$	Trial form for y_p
k	A
$a_mx^m + \dots + a_0$	$A_mx^m + \dots + A_0$
$e^{\alpha x}$	$Ae^{\alpha x}$
$x^me^{\alpha x}$	$(A_mx^m + \dots + A_0)e^{\alpha x}$
$\cos\beta x$ or $\sin\beta x$	$A\cos\beta x + B\sin\beta x$
$e^{\alpha x}\cos\beta x$ or $e^{\alpha x}\sin\beta x$	$e^{\alpha x}(A\cos\beta x + B\sin\beta x)$

If any term in the guessed y_p is already contained in y_c, multiply the whole guess by the smallest power x^s that removes the duplication: $y_p = x^s$(original guess).

3.3.5 Undetermined Coefficients: Annihilator Approach

Definition 3.3.9: Annihilator

A differential operator L with constant coefficients is called an *annihilator* of $f(x)$ if

$$L[f(x)] = 0$$

Basic Annihilators.

$$D^n \text{ annihilates } 1, x, x^2, \dots, x^{n-1}.$$

$$(D-\alpha)^n \text{ annihilates } e^{\alpha x}, xe^{\alpha x}, \dots, x^{n-1}e^{\alpha x}.$$

$$\left[(D-\alpha)^2 + \beta^2\right]^n \text{ annihilates } x^ke^{\alpha x}\cos\beta x \text{ and } x^ke^{\alpha x}\sin\beta x, k = 0, 1, \dots, n-1.$$

Procedure. To solve $L[y] = g(x)$ by annihilators:

1. Solve $L[y] = 0$ to find the complementary function y_c.
2. Find an operator L_1 such that $L_1[g] = 0$.
3. Apply L_1 to obtain the homogeneous equation $L_1 L[y] = 0$.
4. Solve the higher-order homogeneous equation.
5. Remove the terms already in y_c; the remaining terms give the form of y_p.
6. Substitute into the original equation to determine the coefficients.

The annihilator method justifies the guessing method. It is often more useful for finding the correct form of y_p than for doing the final coefficient calculation.

3.3.6 Variation of Parameters

Procedure.

1. Put the equation in standard form $y'' + P(x)y' + Q(x)y = f(x)$.
2. Solve the homogeneous equation to find y_1, y_2.
3. Compute $W = y_1 y_2' - y_2 y_1'$. Compute u_1' and u_2' using $u_1' = -\frac{y_2 f}{W}, u_2' = \frac{y_1 f}{W}$
4. Integrate to find u_1, u_2, then write $y_p = u_1 y_1 + u_2 y_2$.
5. The general solution is $y = y_c + y_p$.

3.3.7 Cauchy–Euler Equations

Definition 3.3.10: Cauchy–Euler Equation

A second-order Cauchy–Euler equation has the form where a, b, c are constants:

$$ax^2 y'' + bxy' + cy = g(x)$$

More generally, an nth-order Cauchy–Euler equation has powers of x matched to the order of the derivative:

$$a_n x^n y^{(n)} + a_{n-1} x^{n-1} y^{(n-1)} + \cdots + a_1 x y' + a_0 y = g(x)$$

Root Cases. On an interval where $x > 0$:

Root type	General solution
$m_1 \neq m_2$ real	$y = c_1 x^{m_1} + c_2 x^{m_2}$
$m_1 = m_2 = m$	$y = c_1 x^m + c_2 x^m \ln x$
$m = \alpha \pm i\beta$	$y = x^\alpha \big(c_1 \cos(\beta \ln x) + c_2 \sin(\beta \ln x)\big)$

Nonhomogeneous Case.

1. Solve the associated homogeneous Cauchy–Euler equation.
2. Divide by the leading coefficient to put the equation in standard form.
3. Use variation of parameters to find y_p.

3.3.8 Solving Systems of Linear Differential Equations by Elimination

Definition 3.3.11: Solution of a System

A solution of a system of differential equations is a set of sufficiently differentiable functions, such as

$$x = \Phi_1(t), \qquad y = \Phi_2(t), \qquad z = \Phi_3(t)$$

that satisfies every equation in the system on a common interval I.

Procedure.

1. Rewrite the system using the differential operator D.
2. Apply operators to the equations so that one unknown can be eliminated.
3. Solve the resulting higher-order equation for one dependent variable.
4. Substitute back into one original equation to find the other variables.
5. Use the original system to relate the arbitrary constants.

3.4 Series Solutions of Linear Equations

3.4.1 Review of Power Series

Definition 3.4.1: Power Series

A *power series centered at a* is an infinite series of the form

$$\sum_{n=0}^{\infty} c_n(x-a)^n = c_0 + c_1(x-a) + c_2(x-a)^2 + \cdots$$

Convergence and Radius of Convergence. A power series may converge for some values of x and diverge for others. The set of all real numbers x for which the series converges is called its *interval of convergence*. There is a number R, called the *radius of convergence*, such that $\sum_{n=0}^{\infty} c_n(x-a)^n$converges for $|x-a| < R$ and diverges for $|x-a| > R$. At the endpoints $x = a \pm R$, convergence must be checked separately.

Ratio Test. For a power series, the ratio test often gives the radius of convergence. If $L = \lim_{n\to\infty} \left|\frac{c_{n+1}(x-a)^{n+1}}{c_n(x-a)^n}\right|$ then the series converges absolutely when $L < 1$ and diverges when $L > 1$. The test is inconclusive when $L = 1$, which usually corresponds to the endpoints.

Definition 3.4.2: Analytic Function

A function f is *analytic at a* if it can be represented by a power series centered at a with $R > 0$:

$$f(x) = \sum_{n=0}^{\infty} c_n(x-a)^n$$

Proposition 3.4.3

If $\sum_{n=0}^{\infty} c_n(x-a)^n = 0$ for all x in some open interval around a, then every coefficient must be zero: $c_n = 0, n = 0, 1, 2, \ldots$. This is called the *identity property* of power series.

3.4.2 Solutions About Ordinary Points

Definition 3.4.4: Ordinary and Singular Points

Consider the homogeneous linear second-order equation $a_2(x)y'' + a_1(x)y' + a_0(x)y = 0$. After division by $a_2(x)$, write it in standard form $y'' + P(x)y' + Q(x)y = 0$. A point $x = x_0$ is an *ordinary point* if both $P(x)$ and $Q(x)$ are analytic at x_0. Otherwise, $x = x_0$ is a *singular point*.

Theorem 3.4.5: Existence of Power Series Solutions

If $x = x_0$ is an ordinary point of $a_2(x)y'' + a_1(x)y' + a_0(x)y = 0$ then the equation has two linearly independent solutions of the form $y = \sum_{n=0}^{\infty} c_n(x-x_0)^n$. Each power series solution converges at least for $|x - x_0| < R$ where R is the distance from x_0 to the nearest singular point.

Procedure. To find a power series solution about an ordinary point:

1. Assume $y = \sum_{n=0}^{\infty} c_n(x-x_0)^n$.
2. Differentiate term by term to find y' and y''. Substitute y, y', y'' into the differential equation. Make all series use the same power of $x - x_0$. Set each coefficient equal to zero. Solve the resulting recurrence relation.
3. Write the solution in terms of the arbitrary constants, usually c_0 and c_1.

3.4.3 Solutions About Singular Points

Definition 3.4.6: Regular and Irregular Singular Points

Suppose $x = x_0$ is a singular point of $y'' + P(x)y' + Q(x)y = 0$. Then $x = x_0$ is a *regular singular point* if

$$p(x) = (x - x_0)P(x), \qquad q(x) = (x - x_0)^2 Q(x)$$

are both analytic at x_0. A singular point that is not regular is called an *irregular singular point*.

Theorem 3.4.7: Frobenius' Theorem

If $x = x_0$ is a regular singular point of $a_2(x)y'' + a_1(x)y' + a_0(x)y = 0$, then there exists at least one solution of the following form where r is a constant determined by the differential equation

$$y = \sum_{n=0}^{\infty} c_n(x-x_0)^{n+r} = (x-x_0)^r \sum_{n=0}^{\infty} c_n(x-x_0)^n, \qquad c_0 \neq 0$$

Indicial Equation. For a regular singular point at $x = 0$, write $y'' + P(x)y' + Q(x)y = 0$, and define

$$p(x) = xP(x) = a_0 + a_1 x + a_2 x^2 + \cdots, \qquad q(x) = x^2 Q(x) = b_0 + b_1 x + b_2 x^2 + \cdots$$

Then the *indicial equation* is $r(r-1)+a_0r+b_0=0$. Its roots are called the *indicial roots*.

Procedure. To solve near a regular singular point $x=0$:

1. Assume a Frobenius solution $y=\sum_{n=0}^{\infty} c_n x^{n+r}$, $c_0 \neq 0$.
2. Compute y' and y''. Substitute into the differential equation. Set the coefficient of the lowest power of x equal to zero. This gives the indicial equation. Solve the recurrence relation for each indicial root.
3. Determine whether one or two linearly independent Frobenius series are obtained.

Three Cases for Indicial Roots. Assume r_1 and r_2 are real indicial roots with $r_1 \geq r_2$.

Case	Form of solutions
$r_1 - r_2 \notin \mathbb{Z}_{>0}$	Two Frobenius series usually exist
$r_1 - r_2 \in \mathbb{Z}_{>0}$	One Frobenius series always exists; the second may contain $\ln x$
$r_1 = r_2$	Second solution contains $\ln x$

More explicitly $y_1=\sum_{n=0}^{\infty} c_n x^{n+r_1}$, $c_0 \neq 0$. The second solution may have the form

$$y_2 = Cy_1 \ln x + \sum_{n=0}^{\infty} b_n x^{n+r_2}, \qquad b_0 \neq 0,$$

where C may be zero in the second case, but is nonzero in the repeated-root case.

3.5 The Laplace Transform

3.5.1 Definition of the Laplace Transform

Definition 3.5.1: Laplace Transform

Let f be defined for $t \geq 0$. The *Laplace transform* of f is

$$\mathcal{L}\{f(t)\} = F(s) = \int_0^\infty e^{-st} f(t)\,dt = \lim_{b\to\infty} \int_0^b e^{-st} f(t)\,dt$$

The variable t is the original variable, while s is the transform variable.

Basic Transform Table. For suitable values of s,

$f(t)$	$\mathcal{L}\{f(t)\} = F(s)$
1	$\frac{1}{s}$
t^n	$\frac{n!}{s^{n+1}}, \quad n = 1, 2, 3, \ldots$
e^{at}	$\frac{1}{s-a}$
$\sin kt$	$\frac{k}{s^2+k^2}$
$\cos kt$	$\frac{s}{s^2+k^2}$
$\sinh kt$	$\frac{k}{s^2-k^2}$
$\cosh kt$	$\frac{s}{s^2-k^2}$

Proposition 3.5.2

The Laplace transform is linear. If $\mathcal{L}\{f\} = F$ and $\mathcal{L}\{g\} = G$, then

$$\mathcal{L}\{\alpha f(t) + \beta g(t)\} = \alpha F(s) + \beta G(s)$$

Definition 3.5.3: Piecewise Continuity and Exponential Order

A function f is *piecewise continuous* on $[0, \infty)$ if, on every finite interval, it is continuous except possibly at finitely many points, and each discontinuity is finite. A function f is of *exponential order* c as $t \to \infty$ if there exist constants $M > 0$, $c > 0$, and $T > 0$ such that

$$|f(t)| \leq Me^{ct}, \qquad t > T$$

Theorem 3.5.4: Existence of the Laplace Transform

If f is piecewise continuous on $[0, \infty)$ and is of exponential order c, then $\mathcal{L}\{f(t)\}$ exists for $s > c$.

3.5.2 Inverse Transforms and Transforms of Derivatives

Definition 3.5.5: Inverse Laplace Transform

If $\mathcal{L}\{f(t)\} = F(s)$, then $f(t)$ is called the *inverse Laplace transform* of $F(s)$, written

$$f(t) = \mathcal{L}^{-1}\{F(s)\}$$

Partial Fractions. For rational functions, first decompose into simpler fractions. Typical patterns are

$$\frac{P(s)}{(s-a)(s-b)} = \frac{A}{s-a} + \frac{B}{s-b}$$

$$\frac{P(s)}{(s-a)^m} = \frac{A_1}{s-a} + \frac{A_2}{(s-a)^2} + \cdots + \frac{A_m}{(s-a)^m}$$

Theorem 3.5.6: Transform of a Derivative

Suppose the needed derivatives of f are continuous or piecewise continuous and of exponential order. If $\mathcal{L}\{f(t)\} = F(s)$, then

$$\mathcal{L}\{f^{(n)}(t)\} = s^n F(s) - s^{n-1} f(0) - s^{n-2} f'(0) - \cdots - f^{(n-1)}(0)$$

$$\mathcal{L}\{f'(t)\} = sF(s) - f(0), \qquad \mathcal{L}\{f''(t)\} = s^2 F(s) - sf(0) - f'(0)$$

Solving Linear IVPs by Laplace Transform. For a constant-coefficient linear initial-value problem:

1. Take the Laplace transform of both sides. Use initial conditions through the derivative formulas.
2. Solve the algebraic equation for $Y(s)$. Decompose $Y(s)$ into recognizable pieces. Apply $\mathcal{L}^{-1}$ to obtain $y(t)$.

3.5.3 Operational Properties

Theorem 3.5.7: First Translation Theorem

If $\mathcal{L}\{f(t)\} = F(s)$, then

$$\mathcal{L}\{e^{at}f(t)\} = F(s-a) \Leftrightarrow \mathcal{L}^{-1}\{F(s-a)\} = e^{at}f(t)$$

Definition 3.5.8: Unit Step Function

The *unit step function* at a is

$$U(t-a) = \begin{cases} 0, & 0 \le t < a, \\ 1, & t \ge a \end{cases}$$

Theorem 3.5.9: Second Translation Theorem

If $\mathcal{L}\{f(t)\} = F(s)$ and $a > 0$, then

$$\mathcal{L}\{U(t-a)f(t-a)\} = e^{-as}F(s) \Leftrightarrow \mathcal{L}^{-1}\{e^{-as}F(s)\} = U(t-a)f(t-a)$$

Alternative Form. For a term of the form $g(t)U(t-a)$, it is often easier to use

$$\mathcal{L}\{g(t)U(t-a)\} = e^{-as}\mathcal{L}\{g(t+a)\}$$

Theorem 3.5.10: Derivatives of Transforms

If $\mathcal{L}\{f(t)\} = F(s)$, then for $n = 1, 2, 3, \ldots$,

$$\mathcal{L}\{t^n f(t)\} = (-1)^n \frac{d^n}{ds^n} F(s)$$

$$\mathcal{L}\{tf(t)\} = -F'(s)$$

Definition 3.5.11: Convolution

If f and g are piecewise continuous on $[0, \infty)$, their *convolution* is

$$(f * g)(t) = \int_0^t f(\tau)g(t-\tau)\, d\tau$$

Basic Properties of Convolution. Convolution behaves like a generalized product:

$$f * g = g * f, \qquad (f * g) * h = f * (g * h), \qquad f * (g + h) = f * g + f * h$$

Theorem 3.5.12: Convolution Theorem

If $\mathcal{L}\{f(t)\} = F(s)$ and $\mathcal{L}\{g(t)\} = G(s)$, then

$$\mathcal{L}\{(f * g)(t)\} = F(s)G(s) \Leftrightarrow \mathcal{L}^{-1}\{F(s)G(s)\} = f * g$$

Transform of an Integral. Taking $g(t) = 1$ in the convolution theorem gives

$$\mathcal{L}\left\{\int_0^t f(\tau)\,d\tau\right\} = \frac{F(s)}{s}, \qquad F(s) = \mathcal{L}\{f(t)\}$$

Definition 3.5.13: Volterra Integral Equation

An equation of the following form is a *Volterra integral equation.* The integral term is a convolution, so Laplace transforms turn the equation into an algebraic equation for $F(s) = \mathcal{L}\{f(t)\}$

$$f(t) = g(t) + \int_0^t f(\tau)h(t-\tau)\,d\tau$$

Theorem 3.5.14: Transform of a Periodic Function

If f is piecewise continuous on $[0, \infty)$ and periodic with period T, then

$$\mathcal{L}\{f(t)\} = \frac{\displaystyle\int_0^T e^{-st} f(t)\,dt}{1 - e^{-sT}}$$

3.5.4 The Dirac Delta Function

Unit Impulse. A unit impulse models a force of very large magnitude acting over a very short time. A rectangular impulse centered near t_0 has total area 1, and as its width tends to zero, it is idealized by the Dirac delta.

Definition 3.5.15: Dirac Delta

The *Dirac delta* $\delta(t - t_0)$ is not an ordinary function. It is a generalized function characterized formally by

$$\delta(t-t_0) = 0 \quad (t \neq t_0), \qquad \int_0^\infty \delta(t-t_0)\,dt = 1$$

Theorem 3.5.16: Transform of the Dirac Delta Function

For $t_0 > 0$,

$$\mathcal{L}\{\delta(t-t_0)\} = e^{-st_0}$$

3.6 Systems of Linear Differential Equations

3.6.1 Theory of Linear Systems

Definition 3.6.1: First-Order System

A system of n first-order differential equations has normal form

$$\frac{dx_i}{dt} = g_i(t, x_1, x_2, \ldots, x_n), \qquad i = 1, 2, \ldots, n$$

If each g_i is linear in the dependent variables, the system is linear.

Matrix Form. If $\mathbf{F}(t) = \mathbf{0}$, the system is *homogeneous*; otherwise it's *nonhomogeneous.*

$$\mathbf{X}' = A(t)\mathbf{X} + \mathbf{F}(t), \qquad \mathbf{X} = \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix}, \qquad \mathbf{F}(t) = \begin{pmatrix} f_1(t) \\ f_2(t) \\ \vdots \\ f_n(t) \end{pmatrix}$$

Any nth-order equation $y^{(n)} = f(t, y, y', \ldots, y^{(n-1)})$ can be rewritten as a first-order system by setting $x_1 = y, x_2 = y', \ldots, x_n = y^{(n-1)}$. Then $x_1' = x_2, x_2' = x_3, \ldots, x_{n-1}' = x_n, x_n' = f(t, x_1, x_2, \ldots, x_n)$

Definition 3.6.2: Solution Vector

A *solution vector* on an interval I is a column vector whose entries are differentiable and satisfy the system on I.

$$\mathbf{X}(t) = \begin{pmatrix} x_1(t) \\ x_2(t) \\ \vdots \\ x_n(t) \end{pmatrix}$$

Theorem 3.6.3: Existence and Uniqueness for Linear Systems

Suppose the entries of $A(t)$ and $\mathbf{F}(t)$ are continuous on an interval I containing t_0. Then the initial-value problem $\mathbf{X}' = A(t)\mathbf{X} + \mathbf{F}(t), \mathbf{X}(t_0) = \mathbf{X}_0$ has a unique solution on I.

Theorem 3.6.4: Superposition for Homogeneous Systems

If $\mathbf{X}_1, \ldots, \mathbf{X}_k$ are solution vectors of $\mathbf{X}' = A(t)\mathbf{X}$ on I, then $\mathbf{X} = c_1\mathbf{X}_1 + \cdots + c_k\mathbf{X}_k$ is also a solution.

Definition 3.6.5: Linear Independence of Solution Vectors

Solution vectors $\mathbf{X}_1, \ldots, \mathbf{X}_k$ are linearly dependent on I if there exist constants $c_1, \ldots, c_k$, not all zero, such that

$$c_1\mathbf{X}_1(t) + \cdots + c_k\mathbf{X}_k(t) = \mathbf{0}$$

for every $t \in I$. Otherwise, they are linearly independent.

Fundamental Matrix. For n linearly independent solutions $\mathbf{X}_1, \ldots, \mathbf{X}_n$ (fundamental set of solutions), the matrix

$$\Phi(t) = \begin{pmatrix} | & | & & | \\ \mathbf{X}_1 & \mathbf{X}_2 & \cdots & \mathbf{X}_n \\ | & | & & | \end{pmatrix}$$

is called a *fundamental matrix.* Its determinant is the Wronskian of the solution vectors. the set of solution vectors is linearly independent on I if and only if the Wronskian $W(X_1, X_2, \cdots, X_n) = |\Phi(t)| \neq 0$ for every t in the interval.

Theorem 3.6.6: General Solution of a Homogeneous Linear System

If $\Phi(t)$ is a fundamental matrix for $\mathbf{X}' = A(t)\mathbf{X}$, then the general solution is $\mathbf{X}(t) = \Phi(t)\mathbf{C}$ where $\mathbf{C}$ is a constant column vector.

Theorem 3.6.7: General Solution of a Nonhomogeneous System

Let $\mathbf{X}_p$ be a particular solution of $\mathbf{X}' = A(t)\mathbf{X} + \mathbf{F}(t)$, and let $\mathbf{X}_c = \Phi(t)\mathbf{C}$ be the general solution of the associated homogeneous system $\mathbf{X}' = A(t)\mathbf{X}$. Then the general solution is $\mathbf{X}(t) = \mathbf{X}_c(t) + \mathbf{X}_p(t)$.

3.6.2 Homogeneous Linear Systems

Procedure To solve $\mathbf{X}' = A\mathbf{X}$ use the following steps:

1. Find the eigenvalues by solving $\det(A - \lambda I) = 0$.
2. For each eigenvalue λ, find eigenvectors by solving $(A - \lambda I)\mathbf{K} = \mathbf{0}$.
3. If A has n linearly independent eigenvectors, form n solutions $\mathbf{K}_i e^{\lambda_i t}$.
4. If an eigenvalue is repeated but has too few eigenvectors, find generalized eigenvectors. If eigenvalues are complex, use the real and imaginary parts of $\mathbf{K}e^{\lambda t}$ to form real-valued solutions.

Distinct Real Eigenvalues distinctrealeigenvalues If A has n distinct real eigenvalues $\lambda_1, \lambda_2, \ldots, \lambda_n$ with corresponding eigenvectors $\mathbf{K}_1, \mathbf{K}_2, \ldots, \mathbf{K}_n$, then the general solution of $\mathbf{X}' = A\mathbf{X}$ is

$$\mathbf{X}(t) = c_1\mathbf{K}_1 e^{\lambda_1 t} + c_2\mathbf{K}_2 e^{\lambda_2 t} + \cdots + c_n\mathbf{K}_n e^{\lambda_n t}$$

Repeated Eigenvalues: Enough Eigenvectors. Suppose λ is an eigenvalue with algebraic multiplicity m. If there are m linearly independent eigenvectors $\mathbf{K}_1, \mathbf{K}_2, \ldots, \mathbf{K}_m$, then the corresponding part of the general solution is

$$c_1\mathbf{K}_1 e^{\lambda t} + c_2\mathbf{K}_2 e^{\lambda t} + \cdots + c_m\mathbf{K}_m e^{\lambda t}$$

Multiplicity Two with One Eigenvector. Suppose λ has algebraic multiplicity 2 but only one eigenvector $\mathbf{K}$. The first solution is $\mathbf{X}_1 = \mathbf{K}e^{\lambda t}$. To find a second solution, solve for a generalized eigenvector $\mathbf{P}$: $(A - \lambda I)\mathbf{P} = \mathbf{K}$. Then a second linearly independent solution is $\mathbf{X}_2 = (t\mathbf{K} + \mathbf{P})e^{\lambda t}$.

Multiplicity Three with One Eigenvector. Suppose λ has algebraic multiplicity 3 but only one eigenvector $\mathbf{K}$. First solve $(A - \lambda I)\mathbf{K} = \mathbf{0}$. Then find generalized eigenvectors $\mathbf{P}$ and $\mathbf{Q}$ satisfying $(A - \lambda I)\mathbf{P} = \mathbf{K}, (A - \lambda I)\mathbf{Q} = \mathbf{P}$. The three linearly independent solutions are

$$\begin{cases} \mathbf{X}_1 = \mathbf{K}e^{\lambda t} \\ \mathbf{X}_2 = (t\mathbf{K} + \mathbf{P})e^{\lambda t} \\ \mathbf{X}_3 = \left(\frac{t^2}{2}\mathbf{K} + t\mathbf{P} + \mathbf{Q}\right)e^{\lambda t} \end{cases}$$

General Repeated Eigenvalue with One Eigenvector Suppose λ has algebraic multiplicity m but only one eigenvector. We first find a generalized eigenvector chain $(A - \lambda I)\mathbf{K}_1 = \mathbf{0}, (A - \lambda I)\mathbf{K}_2 = \mathbf{K}_1, (A - \lambda I)\mathbf{K}_3 = \mathbf{K}_2, \cdots (A - \lambda I)\mathbf{K}_m = \mathbf{K}_{m-1}$. Here $\mathbf{K}_1$ is the ordinary eigenvector, while $\mathbf{K}_2, \ldots, \mathbf{K}_m$ are generalized eigenvectors.

Then m linearly independent solutions are

$$\begin{cases} \mathbf{X}_1 = \mathbf{K}_1 e^{\lambda t} \\ \mathbf{X}_2 = (t\mathbf{K}_1 + \mathbf{K}_2)\, e^{\lambda t} \\ \mathbf{X}_3 = \left(\frac{t^2}{2!}\mathbf{K}_1 + t\mathbf{K}_2 + \mathbf{K}_3\right) e^{\lambda t} \\ \cdots \\ \mathbf{X}_j = \left(\frac{t^{j-1}}{(j-1)!}\mathbf{K}_1 + \frac{t^{j-2}}{(j-2)!}\mathbf{K}_2 + \cdots + t\mathbf{K}_{j-1} + \mathbf{K}_j\right) e^{\lambda t}, \qquad j = 1, 2, \ldots, m \end{cases}$$

Complex Eigenvalues. Suppose A has real entries and has a complex eigenvalue $\lambda = \alpha + i\beta, \beta > 0$. The conjugate $\overline{\lambda} = \alpha - i\beta$ is also an eigenvalue. Let the eigenvector corresponding to $\lambda = \alpha + i\beta$ be $\mathbf{K} = \mathbf{B} + i\mathbf{C}$, where $\mathbf{B}$ is the real part of $\mathbf{K}$ and $\mathbf{C}$ is the imaginary part of $\mathbf{K}$. Then two real linearly independent solutions are

$$\begin{cases} \mathbf{X}_1 = e^{\alpha t}\left(\mathbf{B}\cos\beta t - \mathbf{C}\sin\beta t\right) \\ \mathbf{X}_2 = e^{\alpha t}\left(\mathbf{B}\sin\beta t + \mathbf{C}\cos\beta t\right) \end{cases}$$

Summary of Cases. For $\mathbf{X}' = A\mathbf{X}$ the form of the general solution:

Case	Solution Form
Distinct real eigenvalues	$\mathbf{K}_i e^{\lambda_i t}$
Repeated eigenvalue with enough eigenvectors	$\mathbf{K}_1 e^{\lambda t}, \ldots, \mathbf{K}_m e^{\lambda t}$
Multiplicity two with one eigenvector	$\mathbf{K}e^{\lambda t}, \quad (t\mathbf{K} + \mathbf{P})e^{\lambda t}$
Multiplicity three with one eigenvector	$\mathbf{K}e^{\lambda t}, \quad (t\mathbf{K} + \mathbf{P})e^{\lambda t}, \quad \left(\frac{t^2}{2}\mathbf{K} + t\mathbf{P} + \mathbf{Q}\right) e^{\lambda t}$
General repeated eigenvalue with one eigenvector	$\mathbf{X}_j = \left(\frac{t^{j-1}}{(j-1)!}\mathbf{K}_1 + \frac{t^{j-2}}{(j-2)!}\mathbf{K}_2 + \cdots + t\mathbf{K}_{j-1} + \mathbf{K}_j\right) e^{\lambda t}$
Complex eigenvalues $\alpha \pm i\beta$	$e^{\alpha t}(\mathbf{B}\cos\beta t - \mathbf{C}\sin\beta t), \quad e^{\alpha t}(\mathbf{B}\sin\beta t + \mathbf{C}\cos\beta t)$

3.6.3 Nonhomogeneous Linear Systems

General Form. A nonhomogeneous linear system has the form $\mathbf{X}' = A\mathbf{X} + \mathbf{F}(t)$. Its general solution is $\mathbf{X} = \mathbf{X}_c + \mathbf{X}_p$ where $\mathbf{X}_c$ solves the associated homogeneous system and $\mathbf{X}_p$ is one particular solution.

Undetermined Coefficients. For constant A, the method of undetermined coefficients applies when the entries of $\mathbf{F}(t)$ are finite combinations of constants, polynomials, exponentials, sines, and cosines.

Procedure.

1. Solve the associated homogeneous system $\mathbf{X}' = A\mathbf{X}$.
2. Guess the form of $\mathbf{X}_p$ from the form of $\mathbf{F}(t)$.
3. Substitute $\mathbf{X}_p$ into $\mathbf{X}' = A\mathbf{X} + \mathbf{F}(t)$.
4. Solve the resulting algebraic equations for the unknown coefficients.
5. Write $\mathbf{X} = \mathbf{X}_c + \mathbf{X}_p$.

Variation of Parameters for Systems. Let $\Phi(t)$ be a fundamental matrix for the homogeneous system $\mathbf{X}' = A(t)\mathbf{X}$.

$$\mathbf{X}(t) = \Phi(t)\mathbf{C} + \Phi(t)\int \Phi^{-1}(t)\mathbf{F}(t)\,dt$$

3.6.4 Matrix Exponential

Definition 3.6.8: Matrix Exponential

For an $n \times n$ matrix A, the *matrix exponential* is defined by

$$e^{At} = I + At + \frac{A^2t^2}{2!} + \frac{A^3t^3}{3!} + \cdots = \sum_{k=0}^{\infty} \frac{A^k t^k}{k!}$$

Derivative and Initial Value. The matrix exponential behaves like the scalar exponential:

$$\frac{d}{dt}e^{At} = Ae^{At} = e^{At}A, \qquad e^{A\cdot 0} = I$$

Theorem 3.6.9: Homogeneous Solution Using the Matrix Exponential

The solution of $\mathbf{X}' = A\mathbf{X}, \mathbf{X}(0) = \mathbf{X}_0$ is

$$\mathbf{X}(t) = e^{At}\mathbf{X}_0$$

More generally, the general solution is

$$\mathbf{X}(t) = e^{At}\mathbf{C}$$

Nonhomogeneous Solution. For constant A,$\mathbf{X}' = A\mathbf{X} + \mathbf{F}(t), \mathbf{X}(t_0) = \mathbf{X}_0$, we have

$$\mathbf{X}(t) = e^{A(t-t_0)}\mathbf{X}_0 + \int_{t_0}^{t} e^{A(t-s)}\mathbf{F}(s)\,ds$$

Computing e^{At} by Diagonalization. If $A = PDP^{-1}$, where $D = \mathrm{diag}(\lambda_1, \ldots, \lambda_n)$ then

$$e^{At} = Pe^{Dt}P^{-1}, \qquad e^{Dt} = \mathrm{diag}(e^{\lambda_1 t}, \ldots, e^{\lambda_n t})$$

Computing e^{At} by Laplace Transform.

$$\mathcal{L}\{e^{At}\} = (sI - A)^{-1}, \qquad e^{At} = \mathcal{L}^{-1}\{(sI - A)^{-1}\}$$